

1	Review: set theory	1-1
2	Some number theory	2-1
3	Groups	3-1
4	Elementary properties of groups	4-1
5	Dihedral groups	5-1
6	Order of an element of a group	6-1
7	Subgroups	7-1
8	Direct products	8-1
9	Cyclic groups	9-1
10	Permutation groups	10-1
11	Homomorphisms	11-1
12	Isomorphisms of groups	12-1
13	Cosets and Lagrange theorem	13-1
14	Quotient groups	14-1
15	Group actions	15-1
16	Finite abelian groups	16-1
17	Rings	17-1
18	Integral domains and fields	18-1
19	Ideals and quotient rings	19-1

20 Ring homomorphisms	20-1
21 Rings of polynomials	21-1
22 Unique factorization domains	22-1
23 Prime elements and UFDs	23-1
24 PIDs and UFDs	24-1

Sets

In general sets will be denoted by capital letters: $A, B, C, \dots$

Frequently used sets:

$\emptyset$ = the empty set (i.e. the set that contains no elements)

$\mathbb{N} = \{0, 1, 2, \dots\}$ the set of natural numbers

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ the set of integers

$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ the set of positive integers

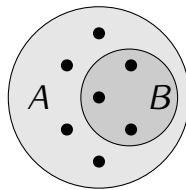
$\mathbb{Q}$ = the set of rational numbers

$\mathbb{R}$ = the set of real numbers

We will write $x \in A$ to denote that x is an element of the set A and $y \notin A$ to indicate that y is not an element of A . For example, $5 \in \mathbb{Z}$, $\frac{1}{3} \notin \mathbb{Z}$.

Subsets

A set B is a *subset* of a set A if every element of B is in A . In such case we write $B \subseteq A$.



A set B is a *proper subset* of A if $B \subseteq A$ and $B \neq A$.

Example. Some subsets of $\mathbb{Z}$:

- $A = \{n \in \mathbb{Z} \mid n > 10\}$ - the set of integers greater than 10.
- $B = \{n \in \mathbb{Z} \mid n = 2k \text{ for some } k \in \mathbb{Z}\}$ - the set of even integers.

Operations on sets

- The *union* of sets A and B is the set $A \cup B$ that consists of all elements that belong to either A or B (or both):

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

- The *intersection* of sets A and B is the set $A \cap B$ that consists of all elements that belong to both A and B :

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

- The *difference* of sets A and B is the set $A \setminus B$ that consists of all elements that belong to A but not to B :

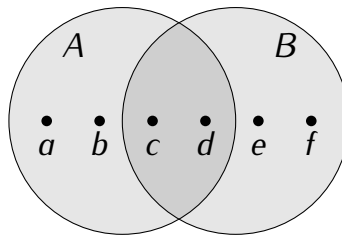
$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example. If $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$ then:

$$A \cup B = \{a, b, c, d, e, f\}$$

$$A \cap B = \{c, d\}$$

$$A \setminus B = \{a, b\}$$



Note. We say that sets A and B are *disjoint sets* if $A \cap B = \emptyset$.

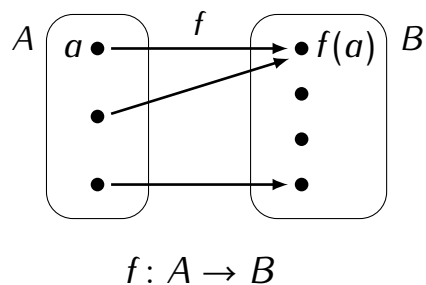
- The *Cartesian product* of sets A , B is the set consisting of all ordered pairs of elements of A and B :

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

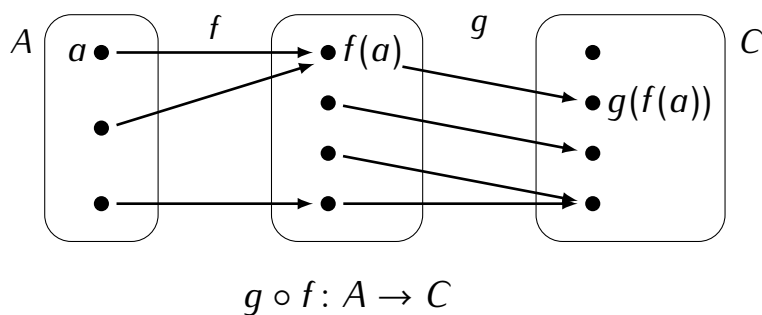
Example. If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$$

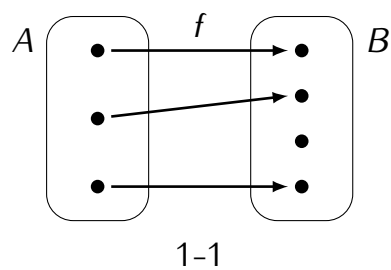
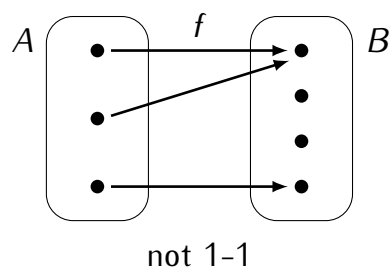
Functions



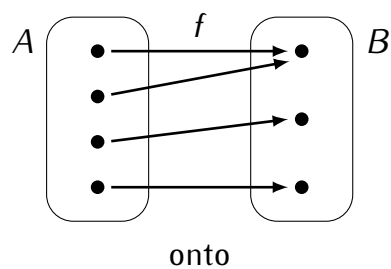
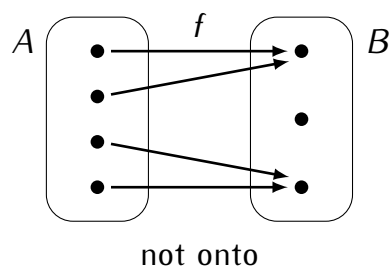
Composition of functions:



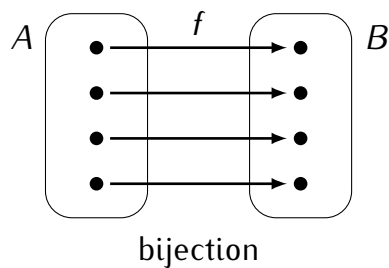
- A function $f: A \rightarrow B$ is *1-1* if $f(a) = f(a')$ only if $a = a'$.



- A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is $a \in A$ such that $f(a) = b$.



- A function $f: A \rightarrow B$ is a *bijection* if f is both 1-1 and onto.



Note. If $f: A \rightarrow B$ is a bijection then the inverse function $f^{-1}: B \rightarrow A$ exists and it is also a bijection. Then for every $a \in A$ and $b \in B$ we have:

$$f^{-1}(f(a)) = a \quad \text{and} \quad f(f^{-1}(b)) = b$$

Cardinality

We will denote by $|A|$ the cardinality of the set A . For a finite set, this is the number of elements of A .

Note. If A, B are sets, then $|A| = |B|$ if and only if there exists a bijection $f: A \rightarrow B$.

Divisibility of integers.

If m, n are integers then we say that m *divides* n if $n = km$ for some integer k . We write: $m|n$.

Some properties of divisibility:

- For every n we have $1|n$.
- For every $m \neq 0$ we have $m|0$
- If $k|m$ and $m|n$ then $k|n$
- If $m|n$ and $n|m$ then either $m = n$ or $m = -n$

The Greatest Common Divisor

The *greatest common divisor* of integers $n_1, \dots, n_k$ is the greatest integer m such that $m|n_i$ for $i = 1, \dots, k$. We denote this integer by $\gcd(n_1, \dots, n_k)$.

Some properties of gcd:

- If $m|n_i$ for $i = 1, \dots, k$ then $m|\gcd(n_1, \dots, n_k)$.
- If $n > 0$ then $\gcd(0, n) = n$
- $\gcd(m, n) = \gcd(m, n - m)$
- If m, n are non-zero integers that there exists integers a, b such that

$$\gcd(m, n) = am + bn$$

Moreover, $\gcd(m, n)$ is the smallest positive integer of such form.

If $\gcd(m, n) = 1$ then we say that m and n are *relatively prime*.

The Least Common Multiple

The *least common multiple* of integers $n_1, \dots, n_k$ is the smallest positive integer m such that $n_i | m$ for $i = 1, \dots, k$. We denote this integer by $\text{lcm}(n_1, \dots, n_k)$.

Some properties of lcm:

- If m is an integer such that $n_i | m$ for $i = 1, \dots, k$ then $\text{lcm}(n_1, \dots, n_k) | m$.
- $\text{lcm}(n_1, \dots, n_k) = \frac{n_1 \cdot \dots \cdot n_k}{\text{gcd}(n_1, \dots, n_k)}$

Prime numbers.

An integer $p > 1$ is prime if the only positive integers dividing p are p and 1.

Some properties of primes:

- If p is a prime and $p | mn$ then either $p | m$ or $p | n$.
- If $n > 1$ is any integer then there is a unique way of writing n as a product of primes:

$$n = p_1 p_2 \cdot \dots \cdot p_k$$

such that $p_1 \geq p_2 \geq \dots \geq p_k$.

Definition 3.1

A *group* is set G equipped with an operation that assigns to each pair of elements $a, b \in G$ an element $a \cdot b \in G$ in such way, that the following conditions are satisfied:

- 1) For any $a, b, c \in G$ we have

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

(associativity).

- 2) There exists an element $e \in G$ such that

$$e \cdot a = a \cdot e = a$$

for all $a \in G$. The element e is called the *identity element* or the *trivial element*.

- 3) For each element $a \in G$ there exists an element $b \in G$ such that

$$a \cdot b = b \cdot a = e$$

Such element b is called the *inverse* of a and it is denoted by a^{-1} .

Note. The first condition in the definition of a group implies that if we want to multiply several elements, then it does not matter how we place parentheses. E.g.:

$$((ab)c)d = (ab)(cd) = a(b(cd)) = a((bc)d) = (a(bc))d$$

Definition 3.2

A *abelian group* is a group G where the multiplication is commutative:

$$a \cdot b = b \cdot a$$

for all $a, b \in G$.

Notation

Multiplicative:

- $a \cdot b, a \times b, a * b, a \circ b, a \odot b, \dots$
- Inverse element: a^{-1} .
- The identity element: $e, 1, \dots$

Additive:

- $a + b$
- Inverse element: $-a$.
- The identity element: 0 .

Note: The additive notation is only used for abelian groups.

Some examples of groups

- $\mathbb{Z}$ – the group of integers (with addition)
- $\mathbb{Q}$ – the group of rational numbers (with addition)
- $\mathbb{R}$ – the group of real numbers (with addition)
- $\mathbb{C}$ – the group of complex numbers (with addition)
- $\mathbb{Q}^*$ – the group of non-zero rational numbers (with multiplication)
- $\mathbb{R}^*$ – the group of non-zero real numbers (with multiplication)
- $\mathbb{C}^*$ – the group of non-zero complex numbers (with multiplication)
- $\mathbb{Q}^+$ – the group of positive rational numbers (with multiplication)
- $\mathbb{R}^+$ – the group of positive real numbers (with multiplication)
- The trivial group $\{e\}$.

Example: General linear groups $GL(n, \mathbb{R})$.

- Elements of $GL(n, \mathbb{R})$: invertible $n \times n$ matrices with real entries.

- **Group operation:** matrix multiplication.
- **The identity element:** the identity matrix I_n .

Example: Groups $\mathbb{Z}_n$

Notation: Let $n > 0$ be an integer. For any integer m we have

$$m = qn + r$$

where $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then we write

$$m \bmod n = r$$

For an integer $n \geq 2$ the group $\mathbb{Z}_n$ is defined as follows:

- **Elements of $\mathbb{Z}_n$:** numbers $0, 1, \dots, n-1$
- **Group operation $\oplus$:** For $k, l \in \mathbb{Z}_n$ we set

$$k \oplus l := (k + l) \bmod n$$

- **The identity element:** 0.
- **Inverses:** The inverse of an element $k \in \mathbb{Z}_n$ is the element $n - k$.

Example: Groups $U(n)$

Recall:

- If m, n are integers then $\gcd(m, n)$, is the greatest integer that divides both m and n .
- For any $m, n \neq 0$ there exists $a, b \in \mathbb{Z}$ such that

$$am + bn = \gcd(m, n)$$

Moreover, $\gcd(m, n)$ is the smallest positive integer that can be obtained for any choice of a and b .

For an integer $n \geq 2$ the group $U(n)$ is defined as follows:

- **Elements of $U(n)$:** integers $1 \leq k < n$ such that $\gcd(k, n) = 1$
- **Group operation $\odot$:** For $k, l \in U(n)$ we set

$$k \odot l := (k \cdot l) \bmod n$$

- **The identity element:** 1.
- **Inverses:** If k is an element of $U(n)$, then we can find $p, q \in \mathbb{Z}$ such that $pk + qn = \gcd(k, n) = 1$. Let $\bar{p} = p \bmod n$. Notice that we have $\gcd(p, n) = 1$, so also $\gcd(\bar{p}, n) = 1$. Also,

$$(\bar{p} \cdot k) \bmod n = (pk) \bmod n = (pk + qn) \bmod n = 1$$

This means that $\bar{p} = k^{-1}$ in the group $U(n)$.

Theorem 4.1

In a group G there is only one identity element.

Proof. If $e, e' \in G$ are two identity elements, then for every $a \in G$ we have

$$a = ea \quad \text{and} \quad ae' = a$$

This gives:

$$e' = ee' = e$$

□

Theorem 4.2 (Cancellation Property)

If G is a group, $a, b, c \in G$ and either $ac = bc$ or $ca = cb$ then $a = b$.

Proof. Assume that $ac = bc$. Then we have

$$a = (ac)c^{-1} = (bc)c^{-1} = b$$

□

Theorem 4.3

If G is a group then every element of G has only one inverse element.

Proof. Let $a \in G$ and let $b, b' \in G$ be two inverses of a :

$$ab = e = ab'$$

Then, by the cancellation property, $b = b'$.

□

Theorem 4.4

If G is a group and $a, b \in G$ then $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. We have:

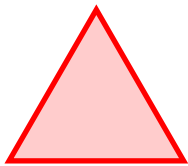
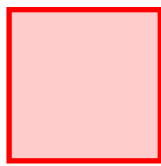
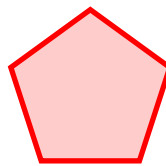
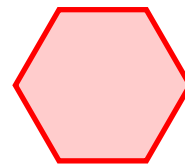
$$(ab)(b^{-1}a^{-1}) = abb^{-1}a^{-1} = e$$

Similarly, $(b^{-1}a^{-1})(ab) = e$. □

Exercise. Write multiplication tables of all possible groups with 1, 2 and 3 elements.

Exercise. Write multiplication tables of all possible groups with 4 elements.

Regular polygons P_n with n sides:

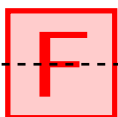
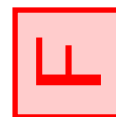
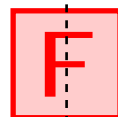
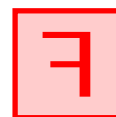
 P_3  P_4  P_5  P_6

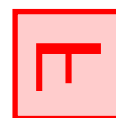
Symmetries of P_4 :


 $\xrightarrow{I}$

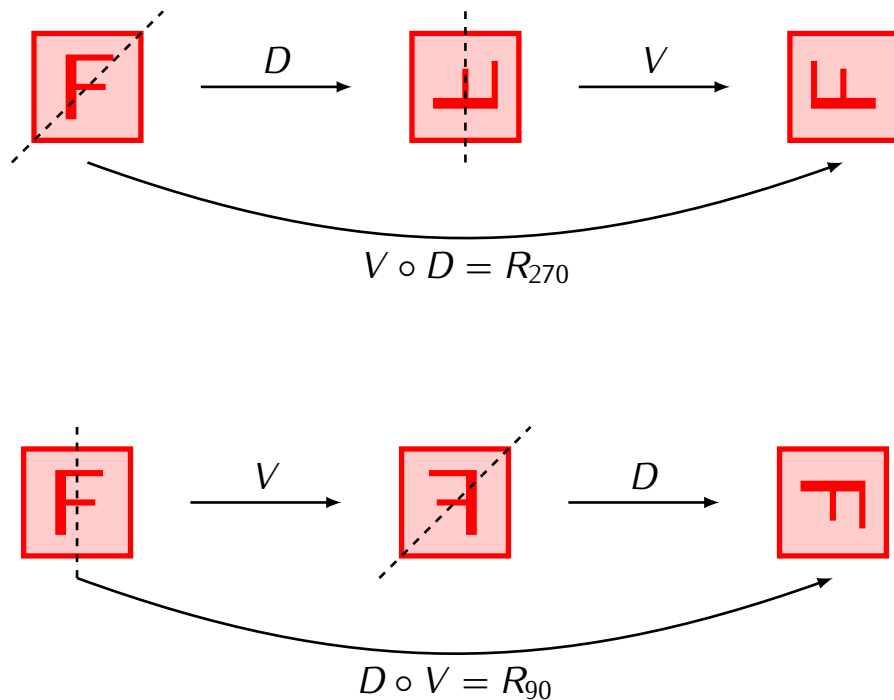
 $\xrightarrow{R_{90}}$

 $\xrightarrow{R_{180}}$

 $\xrightarrow{R_{270}}$

 $\xrightarrow{H}$

 $\xrightarrow{V}$

 $\xrightarrow{D}$

 $\xrightarrow{D'}$


Composition of symmetries:



Composition table of symmetries of a square:

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

For $n \geq 3$ the dihedral group D_n is defined as follows:

- **Elements of D_n :** symmetries of the regular polygon with n sides.
- **Group operation:** Composition of symmetries (e.g. $V \circ D = R_{270}$).
- **The identity element:** The identity symmetry I .

Definition 5.1

The *order* of a group G is the number of elements of G . It is denoted by $|G|$. If G has infinitely many elements, we write $|G| = \infty$.

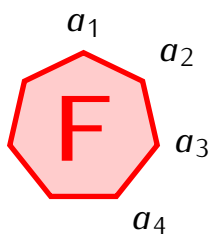
Examples.

- $|D_4| = 8$
- $|\mathbb{Z}_n| = n$
- $|\mathbb{Z}| = \infty$
- $|\mathbb{Q}| = \infty$
- $|\mathbb{R}| = \infty$
- $|GL(n, \mathbb{R})| = \infty$

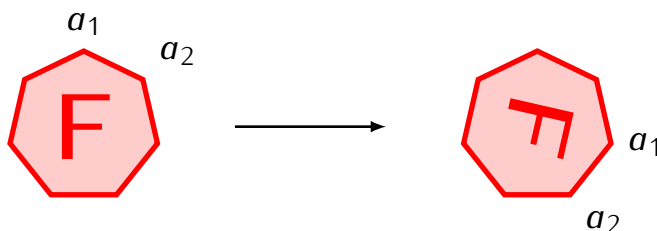
Theorem 5.2

For $n \geq 3$ we have $|D_n| = 2n$.

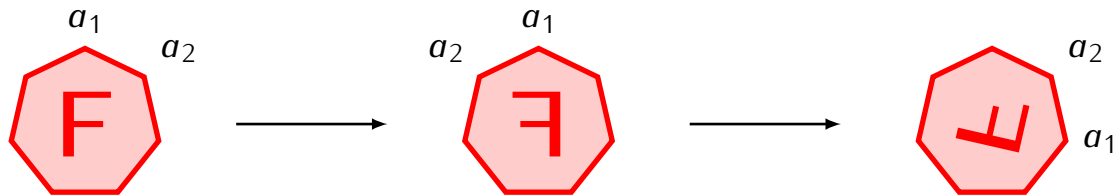
Proof. Let P_n be a regular polygon with vertices $a_1, a_2, \dots, a_n$:



Each symmetry of P_n is uniquely determined once we know where it sends vertices a_1 and a_2 . For every $1 \leq i \leq n$ there is a symmetry that sends a_1 to a_i and a_2 to a_{i+1} given by a rotation:



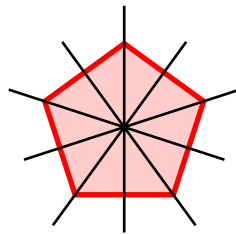
Also, for every i we there is a symmetry that sends a_1 to a_i and a_2 to a_{i-1} . This is given by a composition of a reflection with respect to a line that passes through a_1 and a rotation:



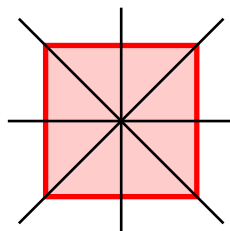
Altogether, this gives $2n$ possible symmetries of P_n . □

Note. The dihedral group D_n consists of the following elements:

- 1) n rotations by the angles of $k \cdot \frac{360}{n}$ degrees for $k = 0, \dots, n - 1$. For $k = 0$ this gives the rotation by 0 degrees, i.e. the identity symmetry.
- 2) n reflections with respect to different symmetry axes. If n is odd, there is one symmetry axis for each vertex of the polygon P_n :



If n is even, there are $\frac{n}{2}$ symmetry axes passing through pairs of opposite vertices and $\frac{n}{2}$ symmetry axes crossing opposite sides of P_n :



Definition 5.3

Let G be a group, and let $S \subseteq G$ be a subset of G . We say that the set S *generates* G if every element of G can be obtained as a product of some elements of S and inverses of elements of S .

Examples.

- Let P_n be a regular polygon with vertices $a_1, \dots, a_n$. In the proof of Theorem 5.2 we have seen that every symmetry of P_n can be obtained by composing some rotation of P_n and a reflection D with respect to the line that passes through the vertex a_1 . Moreover, every rotation of P_n can be obtained by composing some number of times the rotation R by the angle of $\frac{360}{n}$ degrees. Thus, every symmetry of P_n can be obtained as some product of R and D . This means that the set $\{R, D\}$ generates the dihedral group D_n .
- The group of integers $\mathbb{Z}$ is generated by a set $\{1\}$ consisting of single element $1 \in \mathbb{Z}$, since every element of $\mathbb{Z}$ can be obtained by adding some number of times 1 and -1 .
- The group of integers $\mathbb{Z}_n$ is generated by a set $\{1\}$ consisting of single element $1 \in \mathbb{Z}$.
- The set $\{2\}$ generates $\mathbb{Z}_3$, but it does not generate $\mathbb{Z}_4$.

Exponentiation

Let G be a group and let $g \in G$. For an integer $n > 0$ we denote:

- $g^n = \underbrace{g \cdot g \cdot \dots \cdot g}_{n \text{ times}}$ (additive notation: $ng = \underbrace{g + g + \dots + g}_{n \text{ times}}$)
- $g^{-n} = \underbrace{g^{-1} \cdot g^{-1} \cdot \dots \cdot g^{-1}}_{n \text{ times}}$ (additive notation: $(-n)g = \underbrace{-g - g - \dots - g}_{n \text{ times}}$)
- $g^0 = e$ (additive notation: $0g = 0$)

Properties of exponentiation

- $g^{m+n} = g^m \cdot g^n$ (additive notation: $(m+n)g = (mg) + (ng)$)
- $g^{mn} = (g^m)^n$ (additive notation: $(mn)g = m(ng)$)

Definition 6.1

Let G be a group. An *order* of an element $g \in G$ is the smallest integer $n \geq 1$ such that $g^n = e$. We write: $|g| = n$.

If $g^n \neq e$ for all $n \geq 1$ then we say that g is an element of an *infinite order* and we write $|g| = \infty$.

Note. If $g^n = e$ then $g^{-1} = g^{n-1}$. In particular, if $g^2 = e$ then $g = g^{-1}$.

Exercise. Recall that the multiplication table of the dihedral group D_4 is as follows:

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Find the order of every element of D_4 .

Exercise. Find the order of every element in the group $\mathbb{Z}_6$.

Theorem 6.2

If G is a finite group and $g \in G$ then $|g| < \infty$.

Proof. Consider the sequence

$$g^1, g^2, g^3, \dots \subseteq G$$

Since G consists of finitely many elements, we must have $g^m = g^n$ for some $n > m$. This gives

$$\begin{aligned} g^{-m} g^m &= g^{-m} g^n \\ e &= g^{n-m} \end{aligned}$$

Thus $|g| \leq n - m < \infty$. □

Theorem 6.3

If G is a group, $g \in G$ and $n \geq 1$ is an integer such that $g^n = e$, then $|g|$ divides n .

Proof. We have

$$n = |g| \cdot q + r$$

for some integers $q \geq 0$ and $0 \leq r < |g|$. We want to show that $r = 0$. Assume that it is not true. Then we have

$$e = g^n = g^{|g| \cdot q + r} = g^{|g| \cdot q} \cdot g^r = \left(g^{|g|}\right)^q \cdot g^r = e \cdot g^r = g^r$$

We obtain that $g^r = e$. This is however impossible, since $r < |g|$. □

Theorem 6.4

If G is a group, and $a, b \in G$ are elements such that $|a|, |b| < \infty$ and $ab = ba$ then $|ab|$ divides $|a| \cdot |b|$.

Proof. Let $|a| = m$ and $|b| = n$. We have

$$(ab)^{mn} = a^{mn} b^{mn} = (a^m)^n \cdot (b^n)^m = e^n \cdot e^m = e$$

By Theorem 6.3 we get that $|ab|$ divides $mn = |a| \cdot |b|$. □

Example. In the dihedral group D_4 take $a = R_{90}$, $b = R_{180}$. Then $a \cdot b = R_{90} \cdot R_{180} = R_{270}$. We have $|R_{90}| = 4$, $|R_{180}| = 2$, so $|R_{90}| \cdot |R_{180}| = 8$ which is divisible by $|R_{270}| = 4$.

Example. Theorem 6.4 is not true in general if $ab \neq ba$. Take for example a, b to be two different reflections in the dihedral group D_3 . Then $|a| = |b| = 2$, so $|a| \cdot |b| = 4$, but $|ab| = 3$.

As another example, in the group $GL(2, \mathbb{R})$ take matrices

$$A = \begin{bmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then $|A| = 2$ and $|B| = 2$ but $|AB| = \infty$.

Theorem 6.5

If G is a group, and $a \in G$ is element such that $|a| = n < \infty$ then

$$|a^k| = \frac{n}{\gcd(n, k)}$$

Exercise. Compute the order of the element $6 \in \mathbb{Z}_{10}$.

Proof of Theorem 6.5. Let $|a^k| = r$. Then r is the smallest positive integer such that $(a^k)^r = a^{kr} = e$. Using Theorem 6.3 we obtain that r is the smallest positive integer such that $n|kr$. This means that kr is the least common multiple of n and k : $kr = \text{lcm}(n, k)$. This gives:

$$kr = \text{lcm}(n, k) = \frac{nk}{\gcd(n, k)}$$

and so $r = \frac{n}{\gcd(n, k)}$.

□

Definition 7.1

Let G be a group. A subset $H \subseteq G$ is a *subgroup* of G if it is a group under the operation in G .

Examples.

- $\mathbb{Z}$ and $\mathbb{Q}$ are subgroups of $\mathbb{R}$.
- $\mathbb{Z}$ is a subgroup of $\mathbb{Q}$.
- Let $H \subseteq \mathbb{Z}$ be the set of all odd integers. This is not a subgroup of $\mathbb{Z}$ since e.g. $3, 5 \in H$ but $3 + 5 \notin H$.
- Let $H \subset GL(2, \mathbb{R})$ be a set consisting of all invertible matrices with integer entries. Then H is not a subgroup of $GL(2, \mathbb{R})$ since, for example,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in H \quad \text{but} \quad A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \notin H$$

Note. If G is a group, then the largest subgroup of G is the group G itself. The smallest subgroup of G is the group $\{e\}$ consisting of the identity element of G only.

Theorem 7.2

Let G be a group. A subset $H \subseteq G$ is a subgroup of G if and only if the following conditions are satisfied:

- 1) The identity element e belongs to H .
- 2) If $a, b \in H$ then $a \cdot b \in H$.
- 3) If $a \in H$ then $a^{-1} \in H$.

Exercise. The dihedral group D_4 has the following multiplication table:

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Find all subgroups of D_4 .

Definition 7.3

The *center* of a group G is a set $Z(G) \subset G$ consisting of elements that commute with all elements of G :

$$Z(G) = \{g \in G \mid ag = ga \text{ for all } a \in G\}$$

Theorem 7.4

If G is a group then the center $Z(G)$ of G is a subgroup of G .

Proof. 1) For the identity element $e \in G$ we have

$$ea = a = ae$$

for any $a \in G$, so $e \in Z(G)$

2) Assume that $g, h \in Z(G)$. We will show that then $gh \in Z(G)$. Indeed, for any element $a \in G$ we have

$$a(gh) = (ag)h = (ga)h = g(ah) = g(ha) = (gh)a$$

3) Assume that $g \in Z(G)$. We need to show that then $g^{-1} \in Z(G)$. For any $a \in G$ we have

$$ag^{-1} = (ga^{-1})^{-1} = (a^{-1}g)^{-1} = g^{-1}a^{-1}$$

□

Exercise. Find the center of the dihedral group D_4 .

Definition 7.5

Let G a group and let $a \in G$. The *centralizer* of a in G is the set $C(a) \subseteq G$, which consists of all elements of G that commute with a :

$$C(a) = \{g \in G \mid ag = ga\}$$

Exercise. Find the centralizer of the element V in D_4 .

Theorem 7.6

If G is a group and $a \in G$ then the centralizer $C(a)$ of a in G is a subgroup of G .

Proof. Similar as for Theorem 7.4.

□

Definition 7.7

If G is a group and S is a non-empty subset of G , then $\langle S \rangle$ denotes the smallest subgroup of G containing all elements of S :

$$\langle S \rangle = \{a_1^{k_1} \cdot a_2^{k_2} \cdots a_n^{k_n} \mid n \geq 1 \text{ and for each } i \text{ we have } a_i \in S \text{ and } k_i \in \mathbb{Z}\}$$

We say that $\langle S \rangle$ is the *subgroup of G generated by the set S* .

Note. If $a \in G$ then $\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$.

Exercise. Find the subgroup $\langle 2 \rangle$ in $\mathbb{Z}_{10}$.

Exercise. Find the subgroup $\langle 2 \rangle$ in $\mathbb{Z}_9$.

Exercise. Find the subgroup $\langle V, R_{180} \rangle$ in D_4 .

Recall:

- The order of a group G is the number of elements of G . It is denoted by $|G|$.
- The order of an element a of a group G is the smallest integer $n > 0$ such that $a^n = e$. It is denoted by $|a|$.

Theorem 7.8

Let G be a group, let $a \in G$ and let $\langle a \rangle$ be the subgroup of G generated by a . Then

$$|a| = |\langle a \rangle|$$

Proof. Assume that $|a| = n$. We will show that the group $\langle a \rangle$ consists of n distinct elements: $e = a^0, a^1, a^2, \dots, a^{n-1}$ and so $|\langle a \rangle| = n$.

First, notice that all these elements are different. Indeed, if $0 \leq k < l < n$ and $a^k = a^l$ then $0 < l - k < n$ and

$$a^{l-k} = a^l \cdot a^{-k} = a^k \cdot a^{-k} = e.$$

This is impossible since $l - k$ is smaller than the order of a .

Next, we will show that $\langle a \rangle$ does not contain any elements other than $a^0, a^1, \dots, a^{n-1}$. Each element of $\langle a \rangle$ is of the form a^k for some $k \in \mathbb{Z}$. We have $k = qn + r$ for some $q, r \in \mathbb{Z}$, $0 \leq r < n$. Then $a^k = a^r$. $\square$

Definition 8.1

The *direct product* of groups $G_1, \dots, G_n$ is a group $G_1 \times G_2 \times \dots \times G_n$ defined as follows:

- Elements: n -tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$.
- Group operation:

$$(g_1, g_2, \dots, g_n) \cdot (h_1, h_2, \dots, h_n) = (g_1 h_1, g_2 h_2, \dots, g_n h_n)$$

- The identity element: $(e_1, e_2, \dots, e_n)$ where e_i is the identity element in G_i .
- Inverses: $(g_1, g_2, \dots, g_n)^{-1} = (g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$.

Note. We have:

$$|G_1 \times G_2 \times \dots \times G_n| = |G_1| \cdot |G_2| \cdot \dots \cdot |G_n|$$

Example. The groups $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ has 6 elements:

$$(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)$$

The multiplication table in $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is as follows:

$\circ$	$(0, 0)$	$(0, 1)$	$(0, 2)$	$(1, 0)$	$(1, 1)$	$(1, 2)$
$(0, 0)$	$(0, 0)$	$(0, 1)$	$(0, 1)$	$(1, 0)$	$(1, 1)$	$(1, 2)$
$(0, 1)$	$(0, 1)$	$(0, 2)$	$(0, 0)$	$(1, 1)$	$(1, 2)$	$(1, 0)$
$(0, 2)$	$(0, 2)$	$(0, 0)$	$(0, 1)$	$(1, 2)$	$(1, 0)$	$(1, 1)$
$(1, 0)$	$(1, 0)$	$(1, 1)$	$(1, 2)$	$(0, 0)$	$(0, 1)$	$(0, 2)$
$(1, 1)$	$(1, 1)$	$(1, 2)$	$(1, 0)$	$(0, 1)$	$(0, 2)$	$(0, 0)$
$(1, 2)$	$(1, 2)$	$(1, 0)$	$(1, 1)$	$(0, 2)$	$(0, 0)$	$(0, 1)$

Theorem 8.2

The group $G_1 \times \dots \times G_n$ is abelian if and only if each of the groups G_i is abelian.

Proof. If $G_1, \dots, G_n$ are abelian groups, then

$$\begin{aligned}(g_1, \dots, g_n) \cdot (h_1, \dots, h_n) &= (g_1 h_1, \dots, g_n h_n) \\ &= (h_1 g_1, \dots, h_n g_n) = (h_1, \dots, h_n) \cdot (g_1, \dots, g_n)\end{aligned}$$

Conversely, if $G_1 \times \dots \times G_n$ is abelian then for any $g_i, h_i \in G_i$ we have

$$\begin{aligned}(g_1 h_1, \dots, g_n h_n) &= (g_1, \dots, g_n) \cdot (h_1, \dots, h_n) \\ &= (h_1, \dots, h_n) \cdot (g_1, \dots, g_n) = (h_1 g_1, \dots, h_n g_n)\end{aligned}$$

which gives $g_i h_i = h_i g_i$ for $i = 1, \dots, n$. □

Recall:

- The *least common multiple* of integers $n_1, n_2, \dots, n_k \geq 1$ is the smallest positive integer, denoted by $\text{lcm}(n_1, \dots, n_k)$, which is divisible by each of these numbers.
- If $m > 0$ is an integer divisible by $n_1, \dots, n_k$ then m is divisible by $\text{lcm}(n_1, \dots, n_k)$.

Theorem 8.3

For $i = 1, \dots, n$ let $a_i \in G_i$, and let $(a_1, \dots, a_n) \in G_1 \times \dots \times G_n$. Then

$$|(a_1, \dots, a_n)| = \text{lcm}(|a_1|, \dots, |a_n|)$$

Example. Consider the element $(1, 1) \in \mathbb{Z}_2 \times \mathbb{Z}_3$ since $1 \in \mathbb{Z}_2$ is an element of order 2, and $1 \in \mathbb{Z}_3$ is an element of order 3, we obtain that $|(1, 1)| = \text{lcm}(2, 3) = 6$.

Proof of Theorem 8.3. Let $|(a_1, \dots, a_n)| = p$ and $\text{lcm}(|a_1|, \dots, |a_n|) = q$. We have

$$(a_1, \dots, a_n)^q = (a_1^q, \dots, a_n^q) = (e_1, \dots, e_n)$$

The last equality comes from Theorem 6.3, since $|a_i|$ divides q for each i . Using Theorem 6.3 again we obtain that p divides q . On the other hand,

$$(e_1, \dots, e_n) = (a_1, \dots, a_n)^p = (a_1^p, \dots, a_n^p)$$

which gives $e_i = a_i^p$ for each i . Using Theorem 6.3 one more time, we get that $|a_i|$ divides p , and so $q = \text{lcm}(|a_1|, \dots, |a_n|)$ divides p . As a consequence $p = q$. □

Definition 9.1

A group G is cyclic if there is an element $a \in G$ such that

$$G = \{a^n \mid n \in \mathbb{Z}\}$$

or, in other notation, $G = \langle a \rangle$. In such case we say that a is a *generator* of G .

Example. The following groups are cyclic:

- $\mathbb{Z}$
- $\mathbb{Z}_n$ for any $n \geq 1$
- If G is any group and $a \in G$ then $\langle a \rangle$ is a cyclic subgroup of G .

Theorem 9.2

If G is a finite group then G is cyclic if and only if there is an element $a \in G$ such that $|a| = |G|$.

Proof. If G is cyclic then $G = \langle a \rangle$ for some $a \in G$ and then $|G| = |\langle a \rangle| = |a|$. Conversely, if there is $a \in G$ such that $|a| = |G|$, then $\langle a \rangle \subseteq G$ and $|\langle a \rangle| = |G|$, which gives $\langle a \rangle = G$. $\square$

Theorem 9.3

Every subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$, and let H be a subgroup of G . If H contains only the trivial element $e = a^0$ then H is cyclic since $H = \langle e \rangle$. Otherwise, there are some elements $a^n \in H$ with $n > 0$. Let $m > 0$ be the smallest integer such that $a^m \in H$. We will show that $H = \langle a^m \rangle$.

Since $a^m \in H$, thus $(a^m)^k \in H$ for all $k \in \mathbb{Z}$, so $\langle a^m \rangle \subseteq H$.

Conversely, let $a^n \in H$ for some n . Then $n = qm + r$ for some $0 \leq r < m$. This gives

$$a^n = a^{qm+r} = a^{qm} \cdot a^r$$

We have seen already that $a^{-qm} \in H$, so $a^{-qm} \cdot a^n \in H$. However, we have

$$a^{-qm} \cdot a^n = a^{-qm} \cdot a^{qm} \cdot a^r = a^r$$

which means that $a^r \in H$. Since $r < m$, we get that $r = 0$. Therefore $a^n = a^{qm} \in \langle a^m \rangle$. This implies that $H \subseteq \langle a^m \rangle$. $\square$

Theorem 9.4

If G is a finite cyclic group and $H \subseteq G$ is a subgroup then $|H|$ divides $|G|$.

Proof. Let $G = \langle a \rangle$ and let $|G| = |a| = n$. By Theorem 9.3 we have $H = \langle a^m \rangle$ for some m . Then $|H| = |a^m|$ and by Theorem 6.5 $|a^m| = \frac{n}{\gcd(n,m)}$. Therefore $|H|$ divides $|G|$. $\square$

Theorem 9.5

If G is a finite cyclic group and $d > 0$ is an integer that divides $|G|$ then there exists exactly one subgroup $H \subseteq G$ such that $|H| = d$.

Proof. Let $G = \langle a \rangle$ and let $|G| = |a| = n$. Since d divides n we have $n = dm$ for some $m > 0$. We will first show that a subgroup H of order d exists. Take $H = \langle a^m \rangle$. Then

$$|H| = |a^m| = \frac{n}{\gcd(n,m)} = \frac{n}{m} = d$$

Next, $H' \subseteq G$ be some other subgroup of G such that $|H'| = d$. We have $H' = \langle a^k \rangle$ for some $0 < k \leq n$ such that $\gcd(n, k) = m$. Then $m = pk + qn$ for some $p, q \in \mathbb{Z}$. This gives

$$a^m = a^{pk} \cdot a^{qn} = (a^k)^p \in H'$$

and so $H = \langle a^m \rangle \subseteq H'$. Since both groups H and H' consist of d elements, it follows that $H = H'$. $\square$

Theorem 9.6

Let $G = \langle a \rangle$ be a cyclic group of order n . An element a^k is a generator of G (i.e. $\langle a^k \rangle = G$) if and only if $\gcd(n, k) = 1$.

Proof. The group $\langle a^k \rangle$ consists of $\frac{n}{\gcd(n, k)}$ elements. We have $\langle a^k \rangle = G$ if and only if $\frac{n}{\gcd(n, k)} = n$ i.e. $\gcd(k, n) = 1$. $\square$

Exercise. In the group $\mathbb{Z}_{15}$ find all elements a such that a generates $\mathbb{Z}_{15}$

Theorem 9.7

Let $G_1 = \langle a_1 \rangle$ and $G_2 = \langle a_2 \rangle$ be finite cyclic groups. The group $G_1 \times G_2$ is cyclic if and only if $\gcd(|G_1|, |G_2|) = 1$.

Proof. Assume that $\gcd(|G_1|, |G_2|) = 1$. Consider the element $(a_1, a_2) \in G_1 \times G_2$. By Theorem 8.3 we have:

$$|(a_1, a_2)| = \text{lcm}(|a_1|, |a_2|) = \text{lcm}(|G_1|, |G_2|) = \frac{|G_1| \cdot |G_2|}{\gcd(|G_1|, |G_2|)} = |G_1| \cdot |G_2| = |G_1 \times G_2|$$

This shows that $G_1 \times G_2 = \langle (a_1, a_2) \rangle$.

Conversely, assume that $|G_1| = n_1$, $|G_2| = n_2$ and that $\gcd(n_1, n_2) = d > 1$. Let $(b_1, b_2) \in G_1 \times G_2$ be an arbitrary element. Then

$$(b_1, b_2)^{n_1 n_2 / d} = (b_1^{n_1 \cdot (n_2 / d)}, b_2^{n_2 \cdot (n_1 / d)}) = (e, e)$$

This means that $|(b_1, b_2)|$ divides $n_1 n_2 / d$, and so $|(b_1, b_2)| < n_1 n_2 = |G_1 \times G_2|$. $\square$

Example. The group $\mathbb{Z}_2 \times \mathbb{Z}_3$ is a cyclic group generated by the element $(1, 1)$. On the other hand the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic.

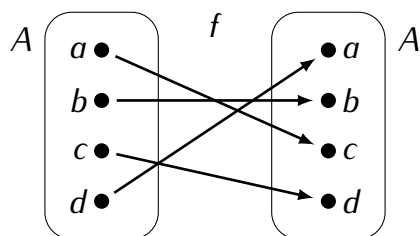
Using induction, Theorem 9.7 can be generalized as follows:

Theorem 9.8

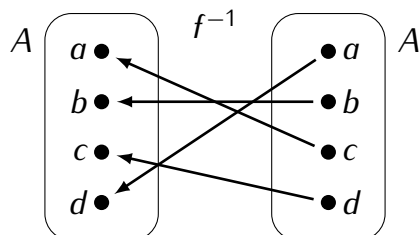
For $i = 1, \dots, n$ let $G_i = \langle a_i \rangle$ be a cyclic group. The group $G_1 \times G_2 \times \dots \times G_n$ is cyclic if and only if $\gcd(|G_i|, |G_j|) = 1$ for all $i \neq j$.

Definition 10.1

A *permutation* of a set A is a function $f: A \rightarrow A$ which is a bijection.



Note. For every permutation f we have the inverse function f^{-1} such that $f \circ f^{-1}(x) = x$ and $f^{-1} \circ f(x) = x$ for all $x \in A$.

**Definition 10.2**

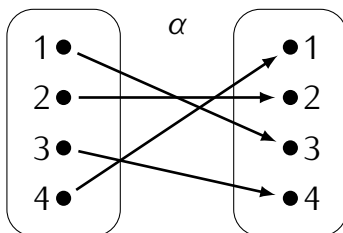
Let A be a set. The *permutation group* of A is a group $S(A)$ defined as follows:

- **Elements of $S(A)$:** permutations $f: A \rightarrow A$.
- **Group operation:** composition of functions $g \circ f$.
- **The identity element:** the function $\varepsilon: A \rightarrow A$, $\varepsilon(x) = x$ for all $x \in A$.
- **The inverse of f :** the inverse permutation f^{-1} .

Definition 10.3

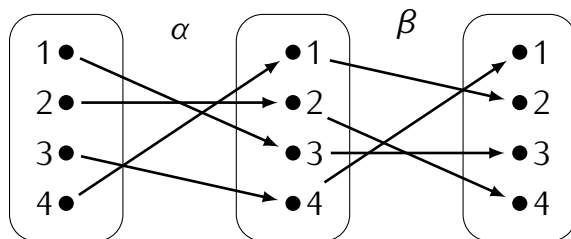
For $n \geq 1$ the group S_n is the group of permutations of the set $A = \{1, 2, \dots, n\}$. This group is called the *symmetric group on n letters*.

Matrix notation of permutations:



$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{bmatrix}$$

Composition:



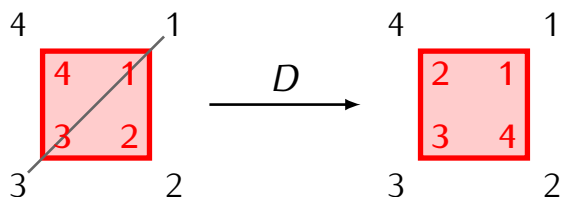
$$\begin{aligned} \beta \circ \alpha &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{bmatrix} \end{aligned}$$

Theorem 10.4

For any $n \geq 1$ we have $|S_n| = n!$

Dihedral groups and permutation groups

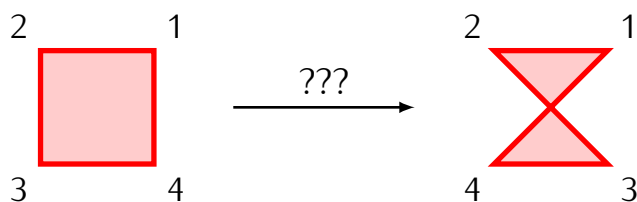
Let P_n be a regular polygon with n vertices. Label the vertices with numbers $1, 2, \dots, n$. Since every symmetry of P_n sends vertices to vertices, it defines a certain permutation of vertices:



$$D = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{bmatrix}$$

Since composition of symmetries corresponds to composition of permutations of vertices, we can identify the dihedral group D_n with a subgroup of the group of permutations S_n . Note that not every permutation in S_n comes from a symmetry of P_n . E.g.:

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{bmatrix}$$



Note. The groups S_n are non-abelian for $n > 2$, e.g:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}^{\alpha} \circ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}^{\beta} &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}^{\alpha \circ \beta} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}^{\beta} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}^{\alpha} &= \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}^{\beta \circ \alpha} \end{aligned}$$

Definition 10.5

Let $\alpha \in S_n$ and let $i \in \{1, \dots, n\}$. We will say that α *moves* i if $\alpha(i) \neq i$. If $\alpha(i) = i$ we will say that α *fixes* i .

Example. The permutation

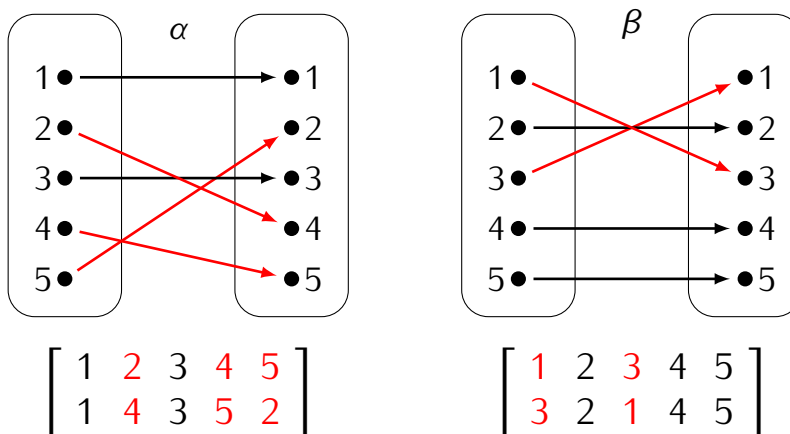
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{bmatrix}$$

moves 1, 2 and 4, and fixes 3.

Definition 10.6

We will say that permutations $\alpha, \beta \in S_n$ are *disjoint* if there is no $i \in \{1, \dots, n\}$ which is moved by both α and β .

Example.



Theorem 10.7

If $\alpha, \beta \in S_n$ are disjoint permutations then

$$\alpha \circ \beta = \beta \circ \alpha$$

Moreover,

$$\alpha \circ \beta(i) = \begin{cases} \alpha(i) & \text{if } i \text{ is moved by } \alpha \\ \beta(i) & \text{if } i \text{ is moved by } \beta \\ i & \text{otherwise} \end{cases}$$

Proof. Assume that $i \in \{1, \dots, n\}$ is an element moved by α . Then α also moves $\alpha(i)$. It follows that both i and $\alpha(i)$ are fixed by β , so we have

$$\beta \circ \alpha(i) = \alpha(i) = \alpha \circ \beta(i)$$

By the same argument, if i is moved by β then

$$\alpha \circ \beta(i) = \beta(i) = \beta \circ \alpha(i)$$

Finally, if both α and β fix i then

$$\alpha \circ \beta(i) = i = \beta \circ \alpha(i)$$

□

Definition 10.8

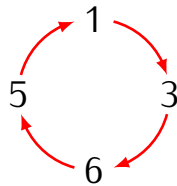
A permutation $\alpha \in S_n$ is a *cycle of length r* (or *r -cycle*) if there are distinct elements $i_1, i_2, \dots, i_r \in \{1, 2, \dots, n\}$ such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and α fixes all other elements of $\{1, \dots, n\}$.

Example.

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 4 & 1 & 5 \end{bmatrix}$$



Cycle notation. A permutation α such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and which fixes all other elements is denoted by $(i_1, i_2, \dots, i_r)$.

Example.

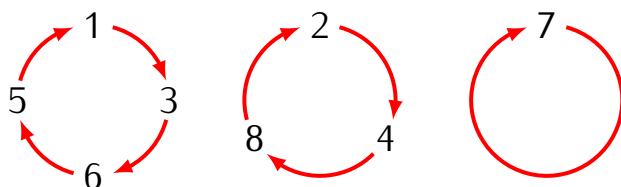
$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 4 & 1 & 5 \end{bmatrix} = (1, 3, 5, 6) = (3, 5, 6, 1) = (5, 6, 1, 3) = (6, 1, 3, 5)$$

Theorem 10.9

Every permutation in S_n is either a cycle or a product of disjoint cycles.

Example.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix} = (1, 3, 6, 5) \circ (2, 8, 4) \circ (7) = (1, 3, 6, 5) \circ (2, 8, 4)$$



Lemma 10.10

Let $\alpha \in S_n$, and let $i_0 \in \{1, \dots, n\}$ be an element moved by α . Then:

- 1) There exists $r > 1$ such that $\alpha^r(i_0) = i_0$
- 2) If $r > 1$ is the smallest integer satisfying $\alpha^r(i_0) = i_0$ then all elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are distinct.

Proof. Consider the sequence

$$i_0 = \alpha^0(i_0), \alpha^1(i_0), \alpha^2(i_0), \dots$$

Since all elements of this sequence come from the finite set $\{1, \dots, n\}$, there must be an integer $r \geq 1$ such that the elements $i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$ are distinct and $\alpha^r(i_0)$ is equal to one of the previous elements. We will show that $\alpha^r(i_0) = i_0$. Indeed, otherwise $\alpha^r(i_0) = \alpha^k(i_0)$ for some $1 \leq k < r$. This gives

$$\alpha(\alpha^{r-1}(i_0)) = \alpha(\alpha^{k-1}(i_0))$$

and since α is a 1-1 function, we obtain

$$\alpha^{r-1}(i_0) = \alpha^{k-1}(i_0)$$

This contradicts the assumption that the elements $i_0, \alpha(i_0), \dots, \alpha^{r-1}(i_0)$ are distinct. $\square$

Proof of Theorem 10.9. Let $\alpha \in S_n$. We will argue that α can be written as a product of cycles by induction with respect to the number k of elements of $\{1, \dots, n\}$ moved by α . If $k = 0$ then α fixes all elements, so it is the identity permutation, which is a 1-cycle.

Assume then that all permutations moving k or fewer elements can be written as a product of disjoint cycles and that α moves $k + 1$ elements. Let $i_0 \in \{1, \dots, n\}$ be an element moved by α . By Lemma 10.10 there is $r > 1$ such that the elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are all distinct and $\alpha^r(i_0) = i_0$. Denote for $k = 1, \dots, r-1$ denote $i_k = \alpha^k(i_0)$. Notice that $\alpha(i_k) = i_{k+1}$ for $k < r-1$ and $\alpha(i_{r-1}) = i_0$. Let $\beta \in S_n$ be a permutation defined as follows:

$$\beta(i) = \begin{cases} i & \text{if } i \in \{i_0, \dots, i_{r-1}\} \\ \alpha(i) & \text{otherwise} \end{cases}$$

Notice that the cycle $(i_0, i_1, \dots, i_{r-1})$ and β are disjoint permutations. Thus, we can use Theorem 10.7 to show that $\alpha = (i_0, i_1, \dots, i_{r-1}) \circ \beta$. Then, since β moves fewer elements than α , by the inductive assumption we can write β as a product of disjoint cycles:

$$\beta = \gamma_1 \circ \dots \circ \gamma_m$$

Therefore we obtain a decomposition of α into a product of disjoint cycles:

$$\alpha = (i_0, i_1, \dots, i_{r-1}) \circ \gamma_1 \circ \dots \circ \gamma_m$$

$\square$

Recall that the least common multiple of integers $n_1, n_2, \dots, n_k \geq 1$ is the smallest positive integer $\text{lcm}(n_1, \dots, n_k)$ which is divisible by each of these numbers.

Theorem 10.11

Assume that a permutation $\alpha \in S_n$ has a decomposition into disjoint cycles

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where γ_i is a cycle of length $r_i > 1$. Then the order of α is given by

$$|\alpha| = \text{lcm}(r_1, r_2, \dots, r_m)$$

Proof. First, notice that if γ is an r -cycle then $|\gamma| = r$. Let

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where γ_i is an r_i -cycle, and let $p = \text{lcm}(r_1, \dots, r_m)$. By Theorem 10.7 disjoint cycles commute, so

$$\alpha^p = (\gamma_1 \circ \cdots \circ \gamma_m)^p = \gamma_1^p \circ \cdots \circ \gamma_m^p = \varepsilon$$

where ε is the identity permutation. By Theorem 6.3 we obtain that $|\alpha|$ divides p .

Next, we claim that $\gamma_i^{|\alpha|} = \varepsilon$ for each i . Indeed, since the cycles are disjoint, elements moved by $\gamma_i^{|\alpha|}$ are fixed by $\gamma_j^{|\alpha|}$ for all $j \neq i$, so if $\gamma_i^{|\alpha|}$ moves some element, then the same element is moved by $\gamma_1^{|\alpha|} \circ \cdots \circ \gamma_m^{|\alpha|}$. This however cannot happen because

$$\gamma_1^{|\alpha|} \circ \cdots \circ \gamma_m^{|\alpha|} = (\gamma_1 \circ \cdots \circ \gamma_m)^{|\alpha|} = \alpha^{|\alpha|} = \varepsilon$$

In this way we obtained that r_i divides $|\alpha|$ for $i = 1, \dots, m$, and so p divides $|\alpha|$. Therefore $|\alpha| = p$. □

Exercise. Compute the order of the following permutation in S_8 :

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix}$$

Exercise. Find all possible orders of elements of S_5 .

Exercise. Compute the number of permutations of order 10 in S_8 .

Exercise. Compute the number of permutations of order 3 in S_7 .

Definition 10.12

A *transposition* in S_n is a cycle (i_1, i_2) of length 2.

Theorem 10.13

Every permutation in S_n can be written as a product of transpositions.

Proof. By Theorem 10.9 every permutation is product of cycles, so it is enough to show that every cycle can be written as a product of transpositions. This is true, since if $(i_1, i_2, \dots, i_r)$ is a cycle in S_n then

$$(i_1, i_2, \dots, i_r) = (i_1, i_r) \circ (i_1, i_{r-1}) \circ \dots \circ (i_1, i_2)$$

□

Note. A permutation can be written as a product of cycles in many different ways:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix} &= \\ &= (1, 3) \circ (1, 2) \\ &= (2, 3) \circ (1, 3) \\ &= (1, 3) \circ (4, 2) \circ (1, 2) \circ (1, 4) \\ &= (2, 4) \circ (1, 2) \circ (2, 3) \circ (1, 4) \\ &= \dots \end{aligned}$$

Theorem 10.14

Let $\alpha \in S_n$ and let

$$\alpha = \beta_1 \circ \beta_2 \circ \dots \circ \beta_r$$

be a decomposition of α into a product of transpositions.

- If the number r is even, then every other decomposition of α into transpositions consists of an even number of transpositions.
- If r is odd, then every other decomposition of α into transpositions consists of an odd number of transpositions.

Lemma 10.15

Let $\beta_1, \dots, \beta_r$ be transpositions in S_n such that

$$\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon$$

where ε is the identity permutation. Then r is an even number.

Proof. We will prove by induction with respect to k the following statement:

For any $k \geq 2$, if $\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon$ and $r \leq k$ then r is an even number.

If $k = 2$ this holds, since the only way to write ε as a product of 1 or 2 transpositions is $\beta \circ \beta^{-1}$, which uses 2 transpositions.

For the inductive step, assume then that the statement holds for some k . We need to show that it also holds for $k + 1$. Let then $\beta_1, \dots, \beta_r$ be transpositions such that $r \leq k + 1$ and

$$\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon \quad (*)$$

Assume that one of the transpositions β_i is of the form (a, b) for some $a, b \in \{1, \dots, n\}$. One can check that the following identities hold:

- $(a, b) \circ (c, d) = (c, d) \circ (a, b)$
- $(a, b) \circ (b, c) = (b, c) \circ (a, c)$

Here a, b, c, d are distinct elements of the set $\{1, \dots, n\}$. These identities say that when multiplying transpositions, we can move the transposition involving a toward the right side without changing the number of transpositions. Using this observation, we can rewrite the equation $(*)$ as follows:

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (a, b_1) \circ (a, b_2) \circ \dots \circ (a, b_k) = \varepsilon \quad (**)$$

where $\gamma_1, \dots, \gamma_{r-k}$ are transpositions that do not involve a . If $b_1 \neq b_i$ for $i = 2, \dots, k$, then the permutation on the left hand side of the equation $(**)$ would send b_1 to a , which is impossible. This means that there is $i > 1$ such that $b_1 \neq b_2, \dots, b_{i-1}$ and $b_1 = b_i$. We will need one more identity:

- $(a, b) \circ (a, c) = (b, c) \circ (a, b)$ for distinct elements a, b, c .

Using this identity, we can bring the equation $(**)$ to the following form:

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (b_1, b_2) \circ \dots \circ (b_1, b_{i-1}) \circ (a, b_1) \circ (a, b_i) \circ (a, b_{i+1}) \circ \dots \circ (a, b_n) = \varepsilon$$

Since $b_1 = b_i$ we have $(a, b_1) \circ (a, b_i) = \varepsilon$, and the above equation becomes

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (b_1, b_2) \circ \dots \circ (b_1, b_{i-1}) \circ (a, b_{i+1}) \circ \dots \circ (a, b_n) = \varepsilon$$

This expresses ε as a product of $r - 2$ transpositions. Since $r \leq k + 1$, thus $r - 2 \leq k$, and so, by the inductive assumption, $r - 2$ must be an even number. Therefore r is an even as well. $\square$

Proof of Theorem 10.14. Assume that a permutation α can be written as a product of transpositions in two different ways:

$$\alpha = \beta_1 \circ \beta_2 \circ \dots \circ \beta_r$$

$$\alpha = \gamma_1 \circ \gamma_2 \circ \dots \circ \gamma_s$$

Then we have

$$\begin{aligned} \varepsilon &= \alpha \circ \alpha^{-1} \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_1 \circ \gamma_2 \circ \dots \circ \gamma_s)^{-1} \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_s^{-1} \circ \gamma_{s-1}^{-1} \circ \dots \circ \gamma_1^{-1}) \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_s \circ \gamma_{s-1} \circ \dots \circ \gamma_1) \end{aligned}$$

This means that ε is a product of $r + s$ transpositions. Since by Lemma 10.15, $r + s$ is an even number, thus either both r and s are even numbers or they are both odd. $\square$

Definition 10.16

A permutation $\alpha \in S_n$ is *even* if it can be written as a product of even number of transpositions and it is *odd* if it can be written as a product of an odd number of transpositions.

Theorem 10.17

The subset of S_n consisting of all even permutations is a subgroup of S_n .

Definition 10.18

The subgroup of S_n consisting of even permutations is called an *alternating group on n letters* and it is denoted by A_n

Theorem 10.19

For $n \geq 2$ the alternating group A_n has order $\frac{n!}{2}$.

Proof. Let B_n be the set of all odd permutations in S_n . Since $|S_n| = n!$, it is enough to show that $|A_n| = |B_n|$, i.e. that there exists a bijection $f: A_n \rightarrow B_n$. Such bijection can be defined by $f(\alpha) = (1, 2) \circ \alpha$. $\square$

Definition 10.20

The *sign* of a permutation $\alpha \in S_n$ is defined as follows:

$$\text{sign}(\alpha) = \begin{cases} +1 & \text{if } \alpha \text{ is even} \\ -1 & \text{if } \alpha \text{ is odd} \end{cases}$$

Note. Recall that for a square matrix A we can compute its determinant $\det A$. The determinant can be defined using permutations and their signs as follows. For a matrix

$$A = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix}$$

we set

$$\det A = \sum_{\alpha \in S_n} \text{sign}(\alpha) \cdot a_{1,\alpha(1)} \cdot a_{2,\alpha(2)} \cdot \cdots \cdot a_{n,\alpha(n)}$$

Definition 11.1

Let G, H be groups. A group homomorphism is a function

$$f: G \rightarrow H$$

which for any $a, b \in G$ satisfies $f(a \cdot b) = f(a) \cdot f(b)$

Theorem 11.2

Let $f: G \rightarrow H$ be a groups homomorphism. Then:

- $f(e_G) = e_H$ where e_G and e_H are the identity elements in G and H , respectively.
- $f(a^{-1}) = f(a)^{-1}$ for any $a \in G$.

Proof. 1) We have

$$f(e_G) = f(e_G \cdot e_G) = f(e_G) \cdot f(e_G)$$

This gives:

$$e_H = f(e_G) \cdot f(e_G)^{-1} = (f(e_G) \cdot f(e_G)) \cdot f(e_G)^{-1} = f(e_G)$$

2) We have

$$e_H = f(e_G) = f(a \cdot a^{-1}) = f(a) \cdot f(a^{-1})$$

which gives:

$$f(a)^{-1} = f(a)^{-1} \cdot e_H = f(a)^{-1} \cdot (f(a) \cdot f(a^{-1})) = f(a^{-1})$$

□

Example. For $n \geq 2$ take the function $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $f(k) = k \bmod n$. Then f is a group homomorphism.

Example. Recall that for $n \geq 1$, the general linear group $GL(n, \mathbb{R})$ is a group that consists of $n \times n$ invertible matrices with matrix multiplication as the group

operation. Recall also that $\mathbb{R}^*$ is the group of non-zero real numbers with multiplication. For an invertible matrix A its determinant is a non-zero number $\det A$. Moreover, $\det AB = (\det A) \cdot (\det B)$. This means that the determinant defines a homomorphism of groups

$$\det: GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$$

Example. Let S_n be the symmetric group on n letters and let $\text{sign}: S_n \rightarrow \mathbb{Z}_2$ be defined by

$$\text{sign}(\alpha) = \begin{cases} 0 & \text{if } \alpha \text{ is an even permutation} \\ 1 & \text{if } \alpha \text{ is an odd permutation} \end{cases}$$

This gives a homomorphism of groups.

Example. For a group G , consider the function $f: G \rightarrow G$ given by $f(a) = a^{-1}$. In general this function is not a homomorphism. For example, take $G = S_3$, the symmetric group on 3 letters, let $\alpha, \beta \in S_3$ be given by $\alpha = (1, 2)$, $\beta = (2, 3)$. Then we have:

$$f(\alpha \circ \beta) = ((1, 2) \circ (2, 3))^{-1} = (2, 3)^{-1} \circ (1, 2)^{-1} = (2, 3) \circ (1, 2) = (1, 3, 2)$$

$$f(\alpha) \circ f(\beta) = (1, 2)^{-1} \circ (2, 3)^{-1} = (1, 2) \circ (2, 3) = (1, 2, 3)$$

and so $f(\alpha \circ \beta) \neq f(\alpha) \circ f(\beta)$

On the other hand, if G is an abelian group then for any $a, b \in G$ we get

$$f(ab) = (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1} = f(a)f(b)$$

Thus for an abelian group $f(a) = a^{-1}$ defines a homomorphism.

Example. Recall that by $\mathbb{R}$ we denote the group of all real numbers with addition and by $\mathbb{R}^*$ the group of non-zero real numbers with multiplication. Define $g: \mathbb{R} \rightarrow \mathbb{R}^*$ by $g(a) = 2^a$. For $a, b \in \mathbb{R}$ we have

$$g(a + b) = 2^{a+b} = 2^a \cdot 2^b = g(a) \cdot g(b)$$

This means that g is a homomorphism of groups.

Example. For any group G and an element $a \in G$ there is exactly one homomorphism $f: \mathbb{Z} \rightarrow G$ such that $f(1) = a$. This homomorphism is given by $f(m) = a^m$.

Example. For any group G the identity function $\text{id}_G: G \rightarrow G$, given by $\text{id}_G(a) = a$ for all $a \in G$ is a homomorphism.

Theorem 11.3

Let $f: G \rightarrow H$ be a homomorphism of groups and let $a \in G$. If $|a| < \infty$ then $|f(a)|$ divides $|a|$.

Proof. If $|a| = n$ then

$$f(a)^n = f(a^n) = f(e_G) = e_H$$

This means that $|f(a)|$ divides n (see Theorem 6.3). $\square$

Example. Let G be a group and let $a \in G$ be an element such that $|a| = n$. Then for each $k = 1, 2, \dots$ there is exactly one homomorphism $f: \mathbb{Z}_{kn} \rightarrow G$ such that $f(1) = a$. This homomorphism is given by $f(m) = a^m$.

Definition 11.4

Let $f: G \rightarrow H$ be a group homomorphism. The *kernel of f* is the subset of G defined by

$$\text{Ker}(f) = \{g \in G \mid f(g) = e\}$$

The *image of f* is the subset of H given by

$$\text{Im}(f) = \{f(g) \mid g \in G\}$$

Theorem 11.5

If $f: G \rightarrow H$ is a homomorphism of groups then $\text{Ker}(f)$ is a subgroup of G and $\text{Im}(f)$ is a subgroup of H .

Proof. Exercise. $\square$

Example. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$, $f(k) = (k \bmod n)$. Then $\text{Im}(f) = \mathbb{Z}_n$ and $\text{Ker}(f) = \{nq \mid q \in \mathbb{Z}\}$. This subgroup of $\mathbb{Z}$ is often denoted by $n\mathbb{Z}$.

Example. Take the determinant homomorphism $\det: GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$. Then $\text{Im}(\det) = \mathbb{R}^*$. Also, $\text{Ker}(\det) = \{A \in GL(n, \mathbb{R}) \mid \det A = 1\}$. This group is called the *special linear group* and it is denoted by $SL(n, \mathbb{R})$.

Example. For the homomorphism $\text{sign}: S_n \rightarrow \mathbb{Z}_2$ we have $\text{Im}(\text{sign}) = \mathbb{Z}_2$ and $\text{Ker}(\text{sign}) = A_n$, where A_n is the alternating group.

Example. Let G be an abelian group and let $f: G \rightarrow G$ be given by $f(a) = a^{-1}$. Then $\text{Im}(f) = G$ and $\text{Ker}(f) = \{e\}$.

Example. Let $g: \mathbb{R} \rightarrow \mathbb{R}^*$, $g(a) = 2^a$. Then $\text{Im}(g) = \mathbb{R}^+$ and $\text{Ker}(g) = \{0\}$.

Theorem 11.6

If $f: G \rightarrow H$ is a homomorphism then $f(a) = f(b)$ if and only if $b = ak$ for some $k \in \text{Ker}(f)$.

Proof. If $f(a) = f(b)$ then

$$e = f(a)^{-1}f(b) = f(a^{-1})f(b) = f(a^{-1}b)$$

so $a^{-1}b \in \text{Ker}(f)$. Taking $k = a^{-1}b$ we then get $ak = a(a^{-1}b) = b$. Conversely, if $k \in \text{Ker}(f)$ then $f(ak) = f(a)f(k) = f(a)e = f(a)$. $\square$

Corollary 11.7

A homomorphism of groups $f: G \rightarrow H$ is 1-1 if and only if $\text{Ker}(f) = \{e\}$.

Corollary 11.8

If $f: G \rightarrow H$ is a homomorphism of groups, and $f(a) = b$ for some $a \in G$, $b \in H$ then

$$f^{-1}(b) = \{ak \mid k \in \text{Ker}(f)\}$$

Note. If G is a group and $H \subseteq G$ is a subgroup, then there exists a homomorphism

$$f: H \rightarrow G$$

such that $\text{Im}(f) = H$. Indeed, we can take $f: H \rightarrow G$, $f(a) = a$.

We will show, however that, in general, not for every subgroup $H \subseteq G$ there is a homomorphism $f: G \rightarrow K$ such that $H = \text{Ker}(f)$.

Theorem 11.9

Let $f: G \rightarrow H$ is a homomorphism of groups then $g \in \text{Ker}(f)$ if and only if for each $a \in G$ we have $aga^{-1} \in \text{Ker}(f)$.

Proof. Since $f(g) = e$ we obtain:

$$f(aga^{-1}) = f(a)f(g)f(a^{-1}) = f(a)ef(a)^{-1} = f(a)f(a)^{-1} = e$$

and so $aga^{-1} \in \text{Ker}(f)$. □

Definition 11.10

Let G be a group. We say that a subgroup $H \subseteq G$ is a *normal subgroup* of G if for any $h \in H$ and $g \in G$ we have $ghg^{-1} \in H$.

We write $H \triangleleft G$ to denote that H is a normal subgroup of G .

Corollary 11.11

If $f: G \rightarrow H$ is a homomorphism of groups then $\text{Ker}(f)$ is a normal subgroup of G .

Proof. This follows from Theorem 11.9. □

Example. G is an abelian group then any subgroup $H \subseteq G$ is normal since for $h \in H$ and $g \in G$ we have

$$ghg^{-1} = gg^{-1}h = h \in H$$

Example. Recall that the alternating group A_n is a subgroup of the symmetric group S_n consisting of all even permutations. This subgroup is normal, since if α is an even permutation and β is any permutation then $\beta \circ \alpha \circ \beta^{-1}$ is an even permutation.

Example. Consider the dihedral group D_4 :

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

The set $\{I, V\}$ is a subgroup of D_4 . However, this is not a normal subgroup since, for example, we have

$$DVD^{-1} = DVD = R_{90}D = H$$

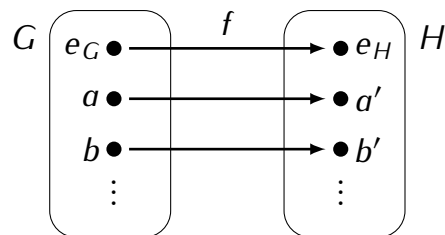
and $H \notin \{I, V\}$.

Exercise. Check that the subgroup of rotations $G = \{I, R_{90}, R_{180}, R_{270}\}$ is a normal subgroup of D_4 .

Note. We will see later that for any normal subgroup $H \triangleleft G$ there is a homomorphism $f: G \rightarrow K$ such that $\text{Ker}(f) = H$.

Definition 12.1

An *isomorphism of groups* is a group homomorphism which is both onto and 1-1.

**Theorem 12.2**

If $g: G \rightarrow H$ is an isomorphism then the inverse function $f^{-1}: H \rightarrow G$ is also an isomorphism.

Proof. The inverse function is 1-1 and onto, so it remains to prove that it is a homomorphism of groups.

Let $a', b' \in H$. We need to show that $f^{-1}(a'b') = f^{-1}(a')f^{-1}(b')$. Since f is a 1-1 function it will suffice to show that $f(f^{-1}(a'b')) = f(f^{-1}(a') \cdot f^{-1}(b'))$.

We have $f(f^{-1}(a'b')) = a'b'$. Also, since f is a homomorphism, we have

$$f(f^{-1}(a')f^{-1}(b')) = f(f^{-1}(a')) \cdot f(f^{-1}(b')) = a'b'$$

□

Theorem 12.3

A homomorphism of groups $f: G \rightarrow H$ is an isomorphism if and only if $\text{Im}(f) = H$ and $\text{Ker}(f) = \{e\}$.

Definition 12.4

We say the group G is *isomorphic* to a group H if there exists an isomorphism $f: G \rightarrow H$. Then we write $G \cong H$.

Theorem 12.5

Isomorphism of groups is an equivalence relation:

- 1) For any group G we have $G \cong G$.
- 2) If G, H are groups such that $G \cong H$ then $H \cong G$.
- 3) If G, H, K are groups such that $G \cong H$ and $H \cong K$, then $G \cong K$.

Proof. 1) The identity function $\text{id}: G \rightarrow G$, $\text{id}(a) = a$ is an isomorphism, so $G \cong G$.

2) If $G \cong H$ then there is an isomorphism $f: G \rightarrow H$. In such case the function $f^{-1}: H \rightarrow G$ is also an isomorphism, so $H \cong G$.

3) If $G \cong H$ and $H \cong K$ then we have isomorphisms $f: G \rightarrow H$ and $g: H \rightarrow K$. Then the composition $g \circ f: G \rightarrow K$ is also an isomorphism and so $G \cong K$. $\square$

Example. Recall that for a set A the group $S(A)$ of permutations of A consists of all permutations $f: A \rightarrow A$ with their composition as the group operation. We will show that if B is another set with the same number of elements as A then $S(A) \cong S(B)$. Indeed, let $\varphi: A \rightarrow B$ be a function which is onto and 1-1. We define

$$\Phi: S(A) \rightarrow S(B)$$

by $\Phi(\alpha) = \varphi \circ \alpha \circ \varphi^{-1}$. One can check that this is an isomorphism of groups.

In particular, recall that the symmetric group on n letters S_n is defined as the group of permutations of the set $\{1, 2, \dots, n\}$. We obtain that if A is a set consists of n elements, then $S(A) \cong S_n$.

Example. If G, H are cyclic groups and $|G| = |H|$ then $G \cong H$. Indeed, let $G = \langle a \rangle$ and $H = \langle b \rangle$ then the function $f: G \rightarrow H$, $f(a^k) = b^k$ is an isomorphism.

As a consequence, we obtain that every cyclic group is isomorphic either to $\mathbb{Z}$ or to $\mathbb{Z}_n$ for some $n \geq 1$.

Example. Let $m, n > 0$ be integers such that $\gcd(m, n) = 1$. Recall that by Theorem 9.7 the group $\mathbb{Z}_m \oplus \mathbb{Z}_n$ is cyclic of order mn . This gives: $\mathbb{Z}_m \oplus \mathbb{Z}_n \cong \mathbb{Z}_{mn}$.

Note. Let $f: G \rightarrow H$ be a homomorphism which is 1-1 (i.e. $\text{Ker}(f) = \{e\}$). Then we can obtain an isomorphism from f by replacing H with $\text{Im}(f)$:

$$f: G \rightarrow \text{Im}(f)$$

Thus any 1-1 homomorphism $f: G \rightarrow H$ defines an isomorphism between G and the subgroup $\text{Im}(f)$ of H .

Theorem 12.6 (Cayley's Theorem)

Let G be a finite group of order n . Then G is isomorphic to a subgroup of the symmetric group S_n .

Proof. Let $\overline{G}$ denote the set of elements of G . Since $\overline{G}$ consists of n elements, the symmetric group S_n is isomorphic to the group $S(\overline{G})$ of permutations of $\overline{G}$. Thus it will suffice to prove that G is isomorphic to a subgroup of $S(\overline{G})$.

To show this, it is enough to construct a 1-1 homomorphism $\Phi: G \rightarrow S(\overline{G})$. We will do it as follows. For an element $a \in G$ define a function $t_a: \overline{G} \rightarrow \overline{G}$ by $t_a(x) = ax$. This function is a permutation of the set $\overline{G}$, so it is an element of the group $S(\overline{G})$. We define: $\Phi(a) = t_a$.

We will show that the function Φ is a group homomorphism, i.e. that for any $a, b \in G$ we have $\Phi(ab) = \Phi(a) \circ \Phi(b)$. Indeed, $\Phi(ab) = t_{ab}$ and $t_{ab}: \overline{G} \rightarrow \overline{G}$ is the permutation given by $t_{ab}(x) = abx$. On the other hand,

$$\Phi(a) \circ \Phi(b) = t_a \circ t_b$$

where $t_a(x) = ax$ and $t_b(x) = bx$. We have

$$t_a \circ t_b(x) = t_a(bx) = abx = t_{ab}(x)$$

Therefore $\Phi(ab) = \Phi(a) \circ \Phi(b)$.

It remains to show that Φ is 1-1, i.e. that $\text{Ker}(\Phi) = \{e\}$. Assume then that $a \in \text{Ker}(\Phi)$. This means that $\Phi(a) = t_a$ is the identity permutation, $t_a(g) = g$ for $g \in G$. But then $ag = g$ for all $g \in G$, which means that $a = e$.

□

Definition 12.7

An *automorphism* of a group G is an isomorphism $f: G \rightarrow G$.

Example. Given a group G and $a \in G$, define $f_a: G \rightarrow G$ by $f_a(g) = aga^{-1}$. This function is an automorphism of G . Automorphisms of this form are called *inner automorphisms*.

Note that if G is an abelian group then G for any $a \in G$ we have $f_a(g) = g$. Thus the only inner automorphism of G is the identity function.

Definition 12.8

Let G be a group. The *group of automorphisms* of G is the group $\text{Aut}(G)$ whose elements are automorphisms of G and the group operation is given by composition of automorphisms.

Definition 13.1

Let G be a group and $H \subseteq G$ a subgroup. For $a \in G$ the *left coset of H in G containing a* is the subset of G given by

$$aH = \{ah \mid h \in H\}$$

Similarly, the *right coset of H in G containing a* is the subset

$$Ha = \{ha \mid h \in H\}$$

Example. Consider the group D_4 :

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Take the subgroup $K = \{I, H\}$ of D_4 . Here are some left and right cosets of K in D_4 :

$$\begin{aligned} R_{90}K &= \{R_{90}, D'\} & KR_{90} &= \{R_{90}, D\} \\ D'K &= \{D', R_{90}\} & KD' &= \{D', R_{270}\} \\ DK &= \{D, R_{270}\} & KD &= \{D, R_{90}\} \end{aligned}$$

Notice that:

- Cosets defined by different elements may be the same. E.g. $R_{90}K = D'K$.
- Left coset of a given element may be different that the right coset. For example, $R_{90}K \neq KR_{90}$.

Theorem 13.2

Let G be a group, $H \subseteq G$ a subgroup, and let $a, b \in G$. Then:

- 1) $a \in aH$.
- 2) either $aH = bH$ or $aH \cap bH = \emptyset$.
- 3) $aH = bH$ if and only if $a^{-1}b \in H$.
- 4) $|aH| = |H|$, where $|aH|$ denotes the number of elements in aH .

Analogous properties hold for right cosets.

Proof.

1) Since $e \in H$ thus $a = ae \in aH$.

2) Assume that $aH \cap bH \neq \emptyset$ and let $g \in aH \cap bH$. Then $ah_1 = g = bh_2$. Then for $h \in H$ we have

$$ah = ah_1(h_1^{-1}h) = bh_2(h_1^{-1}h) \in bH$$

This shows that $aH \subseteq bH$. By a similar argument $bH \subseteq aH$, so $aH = bH$.

3) If $aH = bH$ then $b = ah$ for some $h \in H$, so $a^{-1}b = h \in H$. Conversely, if $a^{-1}b = h \in H$ then $b \in aH \cap bH$. By part 2) this gives $aH = bH$.

4) It is enough to notice that the function $f: H \rightarrow aH$, $f(h) = ah$ is a bijection. $\square$

Definition 13.3

For a group G and a subgroup $H \subseteq G$ by G/H we denote the set of left cosets of H in G and by $H \backslash G$ we denote the set of right cosets.

Example. Cosets of $K = \{I, H\}$ in D_4 :

I	H
R_{90}	D'
R_{180}	V
R_{270}	D

G/H
left cosets

I	H
R_{90}	D
R_{180}	V
R_{270}	D'

$H \backslash G$
right cosets

Theorem 13.4

If G is a group and $H \subseteq G$ is a subgroup, then $|G/H| = |H \backslash G|$.

Proof. The function $f: G/H \rightarrow H \backslash G$ given by $f(aH) = Ha^{-1}$ is a bijection. $\square$

Definition 13.5

If G is a group and $H \subseteq G$ is a subgroup then the *index* of H , denoted $[G : H]$, is the number of left cosets of H in G (or, equivalently, the number of right cosets):

$$[G : H] = |G/H| = |H \backslash G|$$

Example. If $K = \{I, H\} \subseteq D_4$ then $[D_4 : K] = 4$.

Theorem 13.6 (Lagrange Theorem)

If G is a finite group and $H \subseteq G$ is a subgroup then

$$|G| = [G : H] \cdot |H|$$

Proof. By Theorem 13.2 each element of G belongs to exactly one left coset of H . Thus, if $a_1H, a_2H, \dots, a_nH$ are all distinct cosets, then

$$|G| = |a_1H| + |a_2H| + \dots + |a_nH|$$

Moreover, since each coset consists of $|H|$ elements and there are $[G : H]$ cosets, we obtain that $|G| = [G : H] \cdot |H|$. $\square$

Corollary 13.7

If G is a finite group and $H \subseteq G$ is a subgroup then the order of H divides the order of G .

Corollary 13.8

If G is a finite group and $a \in G$ then the order $|a|$ of a divides the order of G .

Proof. Recall that by Theorem 7.8 we have $|a| = |\langle a \rangle|$ where $\langle a \rangle$ is the subgroup of G generated by a . Also, by Corollary 13.7, $|\langle a \rangle|$ divides $|G|$.

□

Note. It is not true that if G is a group and k divides $|G|$ then G contains an element of order k . Take for example the symmetric group S_4 . By looking at possible disjoint cycle decompositions of elements of S_4 , we can see that every element of S_4 has order 1, 2, 3 or 4. This means that S_4 does not contain any element of order 6, even though 6 divides the order of S_4 .

Note. It is also not true that if k divides the order of a group G , then G contains a subgroup H of order k . We will see an example of that later.

Example. Let G be a group of order p where p is a prime number. Then every element of G is of order either 1 (i.e. it is the identity element) or p . Thus if $a \in G$ and $a \neq e$ then $|a| = |G|$. This means that G is a cyclic group generated by a , and so $G \cong \mathbb{Z}_p$.

Example. We will show if G is a group of order 4, then G is isomorphic either to $\mathbb{Z}_4$ or to $\mathbb{Z}_2 \oplus \mathbb{Z}_2$. By Corollary 13.8, if $a \in G$ then $|a| = 1$ (which means that $a = e$), $|a| = 2$, or $|a| = 4$. If G contains an element of order 4, then it is cyclic, and so $G \cong \mathbb{Z}_4$. Otherwise, G contains the trivial element e and three elements, (which we will denote a, b, c) of order 2. Notice that $ab = ba = c$ (since $ab = b$ would give $a = e$, $ab = a$ would give $b = e$, and $ab = e = aa$ would imply that $b = a$). Similarly, we obtain that $ac = ca = b$ and $bc = cb = a$. This shows that the function $f: G \rightarrow \mathbb{Z}_2 \oplus \mathbb{Z}_2$, given by $f(a) = (1, 0)$, $f(b) = (0, 1)$ and $f(c) = (1, 1)$ is an isomorphism of groups.

Recall:

- A normal subgroup of a group G is a subgroup $H \subseteq G$ such that for every $a \in G$ and $h \in H$ we have $aha^{-1} \in H$.
- We write $H \triangleleft G$ to denote that H is a normal subgroup of G .
- If $f: G \rightarrow K$ is a group homomorphism, then $\text{Ker}(f) \triangleleft G$.

Theorem 14.1

Let G be a group and let $H \subseteq G$ be a subgroup. Then the following conditions are equivalent:

- 1) $H \triangleleft G$
- 2) For any $a \in G$ we have $aHa^{-1} = H$ where $aHa^{-1} = \{aha^{-1} \mid h \in H\}$.
- 3) For any $a \in G$ we have $aH = Ha$.

Proof. 1) $\Rightarrow$ 2) Choose $a \in G$. By the definition of a normal subgroup, for every $b \in G$ we have $bHb^{-1} \subseteq H$, so in particular $aHa^{-1} \subseteq H$.

Moreover, taking $b = a^{-1}$ we get $a^{-1}Ha \subseteq H$. This gives:

$$H = eHe^{-1} = (aa^{-1})H(aa^{-1})^{-1} = a(a^{-1}Ha)a^{-1} \subseteq aHa^{-1}$$

Thus we obtain $aHa^{-1} = H$.

2) $\Rightarrow$ 3) Let $a \in G$ and $h \in H$. Since $aHa^{-1} = H$, thus $aha^{-1} = h'$ for some $h' \in H$ and so $ah = h'a \in Ha$. This gives $aH \subseteq Ha$. Similarly, since $a^{-1}Ha = H$, we get $a^{-1}ha = h'$ for some $h' \in H$ and so $ha = ah'$. This implies that $Ha \subseteq aH$.

3) $\Rightarrow$ 1) Let $a \in G$ and $h \in H$. Since $aH = Ha$, there is $h' \in H$ such that $ah = h'a$, or equivalently $aha^{-1} = h'$. Thus $aha^{-1} \in H$ for any $a \in G$ and $h \in H$, which shows that $H \triangleleft G$. $\square$

Theorem 14.2

Let G be a group and $H \triangleleft G$. Let $a_1, a_2, b_1, b_2 \in G$ be elements such that $a_1H = a_2H$ and $b_1H = b_2H$. Then $(a_1b_1)H = (a_2b_2)H$.

Note. Theorem 14.2 is not true if $H \subseteq G$ is a subgroup of G which is not normal. Take, for example, the dihedral group D_4 and let $K = \{I, H\} \subseteq D_4$. We have

$$R_{90}K = D'K \quad \text{and} \quad R_{270}K = DK$$

On the other hand

$$\begin{aligned} (R_{90} \cdot R_{270})K &= IK = \{I, H\} \\ (D' \cdot D)K &= R_{180}K = \{R_{180}, V\} \end{aligned}$$

Thus $(R_{90} \cdot R_{270})K \neq (D' \cdot D)K$.

Proof of Theorem 14.2. Since $a_1H = a_2H$, we have $a_1 = a_2h$ for some $h \in H$. Similarly, since $b_1H = b_2H$, thus $b_1 = b_2h'$ for some $h' \in H$. This gives $a_1b_1 = a_2hb_2h'$.

Since H is a normal subgroup, we have $Hb_2 = b_2H$, so $hb_2 = b_2h''$ for some $h'' \in H$. Using this we obtain

$$a_1b_1 = a_2hb_2h' = a_2b_2h''h' \in a_2b_2H$$

This gives $a_1b_1H \subseteq a_2b_2H$. Analogously we can show that $a_2b_2H \subseteq a_1b_1H$, and so $a_2b_2H = a_1b_1H$. $\square$

Definition 14.3

Let G be a group and let $H \triangleleft G$. The *quotient group* G/H is defined as follows:

- Elements of G/H are left cosets aH of H in G .
- Group operation: $aH \cdot bH = (ab)H$.
- The identity element: the coset $eH = H$.
- The inverse of aH : $a^{-1}H$.

Note. By Lagrange Theorem 13.6 we have

$$|G/H| = [G : H] = \frac{|G|}{|H|}$$

Example. Take the dihedral group D_4 :

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

One can check that the subgroup $K = \{I, R_{180}\}$ is a normal subgroup of D_4 . The quotient group D_4/K has 4 elements:

$$\begin{aligned}
 IK &= R_{180}K = \{I, R_{180}\} \\
 R_{90}K &= R_{270}K = \{R_{90}, R_{270}\} \\
 HK &= VK = \{H, K\} \\
 DK &= D'K = \{D, D'\}
 \end{aligned}$$

The multiplication table of D_4/K is as follows:

$\circ$	IK	$R_{90}K$	HK	DK
IK	IK	$R_{90}K$	HK	DK
$R_{90}K$	$R_{90}K$	IK	DK	HK
HK	HK	DK	IK	$R_{90}K$
DK	DK	HK	$R_{90}K$	IK

Recall that every group of order 4 is isomorphic either to $\mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$. Since all elements of D_4/K are of order 2, we obtain that $D_4/K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Example. If G is a cyclic group every subgroup $H \subseteq G$ is normal, since G is abelian. Moreover, if $G = \langle a \rangle$ then every element of G/H is of the form $a^k H = (aH)^k$ for some $k \in \mathbb{Z}$. This means that G/H is a cyclic group, $G/H = \langle aH \rangle$. If $[G : H] = n$ then $G/H \cong \mathbb{Z}_n$.

Recall that if $f: G \rightarrow H$ is a homomorphism, then by Corollary 11.11 $\text{Ker}(f)$ is a normal subgroup of G , so the quotient group $G/\text{Ker}(f)$ exists.

Theorem 14.4 (First Isomorphism Theorem)

Let $f: G \rightarrow H$ be a homomorphism of groups which is onto. Then

$$H \cong G/\text{Ker}(f)$$

Example. Take the homomorphism $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $f(k) = k \bmod n$. This homomorphism is onto and

$$\text{Ker}(f) = \{k \mid k \bmod n = 0\} = \{nl \mid l \in \mathbb{Z}\}$$

Denote this subgroup of $\mathbb{Z}$ by $n\mathbb{Z}$. By Theorem 14.4 we obtain $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Example. Recall that $\mathbb{R}^*$ denote the group of non-zero real numbers with multiplication. Take the determinant homomorphism

$$\det: GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$$

This homomorphism is onto and

$$\text{Ker}(\det) = \{A \in GL(n, \mathbb{R}) \mid \det A = 1\} = SL(n, \mathbb{R})$$

Thus we obtain $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^*$.

Proof of Theorem 14.4. Let $b \in H$. Since f is onto, there is $a \in G$ such that $f(a) = b$. Recall that by Corollary 11.8 we have

$$f^{-1}(b) = \{ak \mid k \in \text{Ker}(f)\} = a\text{Ker}(f)$$

It follows that we have a well-defined function $\bar{f}: G/\text{Ker}(f) \rightarrow H$ given by $\bar{f}(g\text{Ker}(f)) = f(g)$. We will show that this function is an isomorphism of groups.

First, notice that the function $\bar{f}$ is a homomorphism:

$$\begin{aligned}\bar{f}(a_1\text{Ker}(f) \cdot a_2\text{Ker}(f)) &= \bar{f}(a_1a_2\text{Ker}(f)) \\ &= f(a_1a_2) \\ &= f(a_1) \cdot f(a_2) \\ &= \bar{f}(a_1\text{Ker}(f)) \cdot \bar{f}(a_2\text{Ker}(f))\end{aligned}$$

Next, the function $\bar{f}$ is onto, since f is onto. It remains to show that $\bar{f}$ is 1-1, i.e. that $\text{Ker}(\bar{f}) = \{e\text{Ker}(f)\}$. Assume then that $\bar{f}(g\text{Ker}(f)) = e$. This means that $f(g) = e$, so $g \in \text{Ker}(f)$. But in such case $g\text{Ker}(f) = e\text{Ker}(f)$. $\square$

Corollary 14.5

For any normal subgroup K of a group G there exists a homomorphism $f: G \rightarrow H$ such that $\text{Ker}(f) = K$.

Proof. Take $H = G/K$ and define f by $f(a) = aK$.

□

Definition 15.1

Let G be a group and X be a set. Given a function

$$\mu: G \times X \rightarrow X$$

denote $g \cdot x := \mu(g, x)$. We say that μ is a *group action* of G on the set X if the following conditions are satisfied:

- 1) $(gh) \cdot x = g \cdot (h \cdot x)$ for any $g, h \in G$ and $x \in X$.
- 2) $e \cdot x = x$ for any $x \in X$.

Note. Let $\mu: G \times X \rightarrow X$ be a group action and let $g \in G$. Define a function $\varphi_g: X \rightarrow X$ by $\varphi_g(x) = g \cdot x$. This function is a bijection. Indeed, it is onto, since if $y \in X$, then $y = \varphi_g(g^{-1} \cdot y)$. Also, it is 1-1, since if $\varphi_g(x) = \varphi_g(x')$, then $g \cdot x = g \cdot x'$, which gives

$$x = e \cdot x = g^{-1}g \cdot x = g^{-1}g \cdot x' = e \cdot x' = x'$$

This shows that a group action μ associates to each element $g \in G$ a permutation φ_g of the set X . The property 1) in Definition 15.1 implies that multiplication in G corresponds to composition of permutations:

$$\varphi_{gh} = \varphi_g \circ \varphi_h$$

As a consequence, we obtain a homomorphism of groups:

$$\Phi: G \rightarrow S(X)$$

where $S(X)$ is the group of permutations of the set X and $\Phi(g) = \varphi_g$.

Definition 15.2

Let $\mu: G \times X \rightarrow X$ be a group action.

- The *orbit* of an element $x \in X$ is the subset of X given by

$$\text{Orb}(x) = \{gx \mid g \in G\}$$

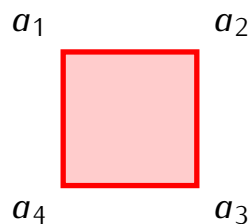
- The *stabilizer* of an element $x \in X$ is the subset of G given by

$$\text{Stab}(x) = \{g \in G \mid gx = x\}$$

Example. Take the dihedral group D_4 :

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Elements of this group are symmetries of a square.



Let $X = \{a_1, a_2, a_3, a_4\}$ be the set of vertices of the square. For each $g \in D_4$ let $g \cdot a_i := g(a_i)$. This defines an action of D_4 on X . Since $a_1 = Ia_1$, $a_2 = R_{90}a_1$, $a_3 = R_{180}a_1$ and $a_4 = R_{270}a_1$, this action has only one orbit:

$$\text{Orb}(a_1) = \{a_1, a_2, a_3, a_4\}$$

The stabilizer of the vertex a_1 consists of elements of D_4 that do not move a_1 . We get $\text{Stab}(a_1) = \{I, D'\}$. On the other hand, $\text{Stab}(a_2) = \{I, D\}$.

Example. Let $G = \langle a \rangle$ be a cyclic group of order 2, so that $a^2 = e$. Define an action of G on the set of integers by $e \cdot n = n$ and $a \cdot n = -n$ for any $n \in \mathbb{Z}$. If $n \neq 0$ then $\text{Orb}(n) = \{n, -n\}$ and $\text{Stab}(n) = \{e\}$. Also, $\text{Orb}(0) = \{0\}$ and $\text{Stab}(0) = \{e, a\}$. Notice that for each $n \in \mathbb{Z}$ we have $\text{Orb}(n) = \text{Orb}(-n)$.

Example. If G is a group then we can define an action of G on itself by $g \cdot x := gxg^{-1}$. Take for example $G = D_4$. We have $\text{Orb}(I) = \{gIg^{-1} \mid g \in D_4\} = \{I\}$ and $\text{Stab}(I) = D_4$. On the other hand $\text{Orb}(R_{90}) = \{R_{90}, R_{270}\}$ and $\text{Stab}(R_{90}) = \{I, R_{90}, R_{180}, R_{270}\}$.

Theorem 15.3

Let $\mu: G \times X \rightarrow X$ be a group action and let $x, y \in X$. Then:

- 1) $x \in \text{Orb}(x)$.
- 2) Either $\text{Orb}(x) = \text{Orb}(y)$ or $\text{Orb}(x) \cap \text{Orb}(y) = \emptyset$.
- 3) $\text{Orb}(x) = \text{Orb}(y)$ if and only if $y = gx$ for some $g \in G$.

Proof.

1) We have $x = e \cdot x \in \text{Orb}(x)$.

2) Assume that $\text{Orb}(x) \cap \text{Orb}(y) \neq \emptyset$ and $z \in \text{Orb}(x) \cap \text{Orb}(y)$. Then $z = g_1 \cdot x$ and $z = g_2 \cdot y$ for some $g_1, g_2 \in G$. Then, for any $h \in G$ we obtain:

$$h \cdot x = (hg_1^{-1}) \cdot (g_1x) = (hg_1^{-1}) \cdot (g_2y) \in \text{Orb}(y)$$

and so $\text{Orb}(x) \subseteq \text{Orb}(y)$. By the same argument $\text{Orb}(y) \subseteq \text{Orb}(x)$, and so $\text{Orb}(x) = \text{Orb}(y)$.

3) If $y = g \cdot x$ then $y \in \text{Orb}(y) \cap \text{Orb}(x)$, and so $\text{Orb}(x) = \text{Orb}(y)$ by 2). Conversely, if $\text{Orb}(x) = \text{Orb}(y)$ then $y \in \text{Orb}(x)$, so $y = g \cdot x$ for some $g \in G$. $\square$

Corollary 15.4

If $\mu: G \times X \rightarrow X$ is a group action and X is a finite set, then

$$|X| = |\text{Orb}(x_1)| + |\text{Orb}(x_2)| + \cdots + |\text{Orb}(x_m)|$$

where $\text{Orb}(x_1), \text{Orb}(x_2), \dots, \text{Orb}(x_m)$ are all different orbits of the action.

Theorem 15.5

Let $\mu: G \times X \rightarrow X$ be a group action and let $x \in X$.

- 1) $\text{Stab}(x)$ is a subgroup of G .
- 2) If $y = gx$ then $\text{Stab}(y) = g \text{Stab}(x) g^{-1}$.
- 3) If G is a finite group then $|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|$

Proof.

1) Since $ex = x$, so $e \in \text{Stab}(x)$. If $g, h \in \text{Stab}(x)$ then $(gh)x = g(hx) = gx = x$, so $gh \in \text{Stab}(x)$. Finally, if $g \in \text{Stab}(x)$ then $g^{-1}x = g^{-1}gx = ex = x$, which gives $g^{-1} \in \text{Stab}(x)$.

2) Let $h \in \text{Stab}(x)$. Then

$$(ghg^{-1})y = ghg^{-1}gx = ghx = gx = y$$

so $g \text{Stab}(x) g^{-1} \subseteq \text{Stab}(y)$. Since $x = g^{-1}y$, this also gives $g^{-1} \text{Stab}(y) g \subseteq \text{Stab}(x)$, or equivalently $\text{Stab}(y) \subseteq g \text{Stab}(x) g^{-1}$. Therefore we obtain $\text{Stab}(y) = g \text{Stab}(x) g^{-1}$.

3) Take the set of left cosets $G/\text{Stab}(x)$. Let $f: G/\text{Stab}(x) \rightarrow \text{Orb}(x)$ be a function given by $f(g \text{Stab}(x)) = gx$. Notice that f is well defined: if $g \text{Stab}(x) = h \text{Stab}(x)$ then $h = ga$ for some $a \in \text{Stab}(x)$, so $hx = gax = gx$. Next, we show that the function f is 1-1. If $f(g \text{Stab}(x)) = f(h \text{Stab}(x))$ then $gx = hx$ so $g^{-1}h \in \text{Stab}(x)$, which means that $g \text{Stab}(x) = h \text{Stab}(x)$. Also, f is onto, since if $y \in \text{Orb}(x)$ then $y = gx = f(g \text{Stab}(x))$.

This shows that f is a bijection and so $|G/\text{Stab}(x)| = |\text{Orb}(x)|$. By Lagrange Theorem 13.6 we obtain

$$|G| = |G/\text{Stab}(x)| \cdot |\text{Stab}(x)| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|.$$

□

Theorem 15.6 (Cauchy Theorem)

If G is a finite group and p is a prime that divides $|G|$ then there exists an element of order p in G .

Proof. Take X to be the set of all p -tuples of elements of G such the product of the p -tuple is the identity element:

$$X = \{(g_0, g_1, \dots, g_{p-1}) \mid g_0 g_1 \cdot \dots \cdot g_{p-1} = e\}$$

Notice that $|X| = |G|^{p-1}$, since in a tuple $(g_0, g_1, \dots, g_{p-1})$ the elements $g_1, g_2, \dots, g_{p-1}$ are arbitrary and $g_0 = (g_1 g_2 \cdot \dots \cdot g_{p-1})^{-1}$. Define an action of $\mathbb{Z}_p$ on X by

$$k \cdot (g_0, g_1, \dots, g_{p-1}) = (g_{0+k}, g_{1+k}, \dots, g_{(p-1)+k})$$

where addition of indices is taken mod p (exercise: check that if $(g_0, g_1, \dots, g_{p-1}) \in X$ then $(g_{0+k}, g_{1+k}, \dots, g_{(p-1)+k}) \in X$). By Theorem 15.5 every orbit of this action divides $|\mathbb{Z}_p| = p$, so it must contain either 1 or p elements. Notice also that $|\text{Orb}((g_0, g_1, \dots, g_{p-1}))| = 1$ if and only if $g_0 = g_1 = \dots = g_{p-1}$. One such orbit is the orbit of the tuple $(e, e, \dots, e)$. If all other orbits consisted of p elements, then the set X would consists of $pq + 1$ elements for some q . This is however impossible, since $|X| = |G|^{p-1}$ is divisible by p . This means that there is some element $g \neq e$ such that $(g, g, \dots, g) \in X$. Then $g^p = e$, and so $|g| = p$. $\square$

Definition 15.7

If p is a prime number then a p -group is a finite group of order p^r for some $r \geq 0$.

Corollary 15.8

A finite group G is a p -group if and only if the order of every element of G is a power of p

Proof. Let G be a p -group, $|G| = p^r$. If $g \in G$ then $|g|$ divides p^r so $|g| = p^i$ for some i .

Conversely, if G is not a p -group then there is some prime $q \neq p$ which divides $|G|$. Then by Theorem 15.6 G contains an element of order q . $\square$

Theorem 15.9

If G is a p -group then there exists an element $a \in G$ such that $a \neq e$ and $ag = ga$ for all $g \in G$.

Proof. Let $|G| = p^r$. Define an action of G on itself by $g \cdot x = gxg^{-1}$. By Theorem 15.5 every orbit $\text{Orb}(x)$ of this action divides p^r , so $|\text{Orb}(x)| = p^i$ for some $i \geq 0$. Also, $|\text{Orb}(x)| = 1$ (i.e. $\text{Orb}(x) = \{x\}$) if and only if $gx = xg$ for all $g \in G$. We have $\text{Orb}(e) = \{e\}$. If all other orbits have more than one element, then we would have $|G| = pq + 1$ for some $q \geq 0$. This is impossible since p divides $|G|$. Therefore there exists some element $a \neq e$ such that $\text{Orb}(a) = \{a\}$. $\square$

The main goal of this section is to prove the following fact:

Theorem 16.1

If G is a finite abelian group then G is isomorphic to a direct product of cyclic groups whose orders are powers of primes:

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \dots \times \mathbb{Z}_{p_k^{r_k}}$$

for primes $p_1, \dots, p_k$ and integers $r_1, \dots, r_k \geq 1$ such that $p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_k^{r_k} = |G|$.

Example. Take $72 = 2^3 \cdot 3^2$. Theorem 16.1 says that every abelian group of order 72 is isomorphic to one of the following groups:

$$\begin{array}{ll} \mathbb{Z}_8 \times \mathbb{Z}_9 & \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \end{array}$$

Definition 16.2

A *short exact sequence* of groups is a sequence group homomorphisms

$$K \xrightarrow{i} G \xrightarrow{q} H$$

such that:

- i is 1-1
- q is onto
- $\text{Im}(i) = \text{Ker}(q)$

Example. Let K be a normal subgroup of G and let $i: K \rightarrow G$ be the inclusion homomorphism: $i(a) = a$. Also, let $q: G \rightarrow G/K$ be the quotient homomorphism $q(a) = aK$. This defines a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} G/K$$

Example. If $f: G \rightarrow H$ is homomorphism which is onto, then we have a short exact sequence

$$\text{Ker}(f) \xrightarrow{i} G \xrightarrow{f} H$$

where $i: \text{Ker}(f) \rightarrow G$ is the inclusion homomorphism.

Example. Let G, H be groups. Define $i: G \rightarrow G \times H$ by $i(g) = (g, e)$, and $q: G \times H \rightarrow H$ by $q(g, h) = h$. This defines a short exact sequence

$$G \xrightarrow{i} G \times H \xrightarrow{q} H$$

Notice that in this case we also have a homomorphism $s: H \rightarrow H \times G$, $s(h) = (e, h)$ and $q \circ s = \text{id}_H$.

Theorem 16.3

Consider a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} H$$

where K, G, H are abelian groups. Assume that there exists a homomorphism $s: H \rightarrow G$ such that $q \circ s(h) = h$ for all $h \in H$. Then $G \cong K \times H$.

Note. Theorem 16.3 is not true in general for non-abelian groups. For example, in the dihedral group D_4 take the subgroup of rotations $K = \{I, R_{90}, R_{180}, R_{270}\}$. Since K is a normal subgroup of D_4 , this gives a short exact sequence

$$K \xrightarrow{i} D_4 \xrightarrow{q} D_4/K$$

The group D_4/K consists of two cosets: IK and VK . Define $s: D_4/K \rightarrow D_4$ by $s(IK) = I$ and $s(VK) = V$. One can check that this is a group homomorphism. Moreover, $q(s(IK)) = IK$ and $q(s(VK)) = VK$. However, D_4 is not isomorphic to $K \times D_4/K$. Indeed, since $K \cong \mathbb{Z}_4$ and $D_4/K \cong \mathbb{Z}_2$, thus $K \times D_4/K \cong \mathbb{Z}_4 \times \mathbb{Z}_2$ is an abelian group, while D_4 is non-abelian.

Proof of Theorem 16.3. Define a function $f: K \times H \rightarrow G$ by $f(k, h) = i(k) \cdot s(h)$. We will show that this function is an isomorphism of groups.

First, we check that f is a homomorphism:

$$\begin{aligned}
 f((k, h) \cdot (k', h')) &= f(kk', hh') \\
 &= i(kk') \cdot s(hh') \\
 &= i(k) \cdot i(k') \cdot s(h) \cdot s(h') \\
 &= (i(k) \cdot s(h)) \cdot (i(k') \cdot s(h')) \\
 &= f(k, h) \cdot f(k', h')
 \end{aligned}$$

Next, assume that $(k, h) \in \text{Ker}(f)$. Then $f(k, h) = i(k) \cdot s(h) = e$. This gives:

$$e = q(i(k) \cdot s(h)) = q(i(k)) \cdot q(s(h)) = e \cdot h = h$$

so $e = h$. Thus, $e = f(k, h) = f(k, e) = i(k) \cdot e = i(k)$. Since i is 1-1, we get that $k = e$. Therefore the only element in $\text{Ker}(f)$ is the identity element (e, e) , which means that f is 1-1.

It remains to show that f is onto. Take an element $g \in G$, and let $h = q(g)$. We have

$$q(gs(h)^{-1}) = q(g) \cdot q(s(q(g)^{-1})) = q(g) \cdot q(g^{-1}) = e$$

which shows that $gs(h)^{-1} \in \text{Ker}(q)$. By exactness, we have $\text{Ker}(q) = \text{Im}(i)$, so there is an element $k \in K$ such that $i(k) = g \cdot s(h)^{-1}$. Consider the element $(k, h) \in K \times H$. We have:

$$f(k, h) = i(k) \cdot s(h) = g \cdot s(h)^{-1} \cdot s(h) = g$$

□

Corollary 16.4

Consider a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} H$$

where K, G, H are abelian groups. Assume that there exists a homomorphism $s: H \rightarrow G$ such that $q \circ s$ is an isomorphism. Then $G \cong K \times H$.

Proof. Define $f = (q \circ s)^{-1} \circ q: G \rightarrow H$. The sequence

$$K \xrightarrow{i} G \xrightarrow{f} H$$

is a short exact sequence. Moreover, we have $f \circ s = (q \circ s)^{-1} \circ q \circ s = \text{id}_H$. Thus by Theorem 16.3 we get $G \cong K \times H$. □

Theorem 16.5

Let G be a finite abelian group. Assume that $|G| = p^r m$ where p is a prime, $r \geq 1$ and m is a number which is not divisible by p . Then $G = K \times H$ where $|K| = p^r$ and $|H| = m$.

Lemma 16.6

Let G be a finite abelian group. Assume that there exists a prime p such that the order of each element $g \in G$ is a power of p . For $m \in \mathbb{Z}$ consider the function

$$f: G \rightarrow G$$

given by $f(g) = g^m$. If m is not divisible by p then f is a group isomorphism.

Proof. By Corollary 15.8 we have $|G| = p^r$ for some $r \geq 0$. Therefore $g^{p^r} = e$ for all $g \in G$.

Since $\gcd(m, p^r) = 1$, there exist $k, l \in \mathbb{Z}$ such that $km + lp^r = 1$. Define $s: G \rightarrow G$ by $s(g) = g^k$. We have

$$s \circ f(g) = (g^m)^k = g^{km} = g^{1-lp^r} = g \cdot g^{-lp^r} = ge = g$$

so $s \circ f = \text{id}_G$. Similarly, $f \circ s = \text{id}_G$. Thus f is an isomorphism and $f^{-1} = s$. $\square$

Proof of Theorem 16.5. Define

$$H := \{g \in G \mid |g| = p^i \text{ for some } i \geq 1\}$$

One can check that this is a subgroup of G . Notice that if $g \in G$ then $g^m \in H$. Indeed, we have

$$(g^m)^{p^r} = g^{mp^r} = g^{|G|} = e$$

so $|g^m|$ divides p^r . As a consequence we obtain a homomorphism $q: G \rightarrow H$, $q(g) = g^m$. This homomorphism is onto since, by Lemma 16.6, $q|_H: H \rightarrow H$ is an isomorphism. Take the short exact sequence

$$\text{Ker}(q) \longrightarrow G \xrightarrow{q} H$$

Define $s: H \rightarrow G$ by $s(h) = h$. The composition $q \circ s: H \rightarrow H$ is given by $q \circ s(h) = h^m$ which is an isomorphism by Lemma 16.6. Using Corollary 16.4 we obtain that $G \cong \text{Ker}(q) \times H$.

We have $|H| \cdot |\text{Ker}(q)| = |G| = p^r m$. By Corollary 15.8, $|H|$ is a power of p . Also, since $\text{Ker}(q) = \{g \in G \mid g^m = 1\}$, thus the order of every element of $\text{Ker}(q)$ divides m . This implies that $\text{Ker}(q)$ does not contain any elements of order p . By Cauchy Theorem 15.6, we obtain that $|\text{Ker}(q)|$ is not divisible by p . Therefore $|H| = p^r$ and $|\text{Ker}(q)| = m$. $\square$

Corollary 16.7

If G is a finite abelian group and $|G| = p_1^{r_1} p_2^{r_2} \cdot \dots \cdot p_k^{r_k}$ where $p_1, p_2, \dots, p_k$ are distinct primes then

$$G = G_1 \times G_2 \times \dots \times G_k$$

where $|G_i| = p_i^{r_i}$.

Proof. We use induction with respect to the number of distinct primes k . If $k = 1$ then $|G| = p_1^{r_1}$, so we take $G_1 = G$.

Assume that the statement is true for all abelian groups whose order is a product of powers of k distinct primes, and let G be a group such that

$$|G| = p_1^{r_1} p_2^{r_2} \cdot \dots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$$

where $p_1, \dots, p_{k+1}$ are distinct primes. By Theorem 16.5 we get $G = G_1 \times H$ where $|G_1| = p_1^{r_1}$ and $|H| = p_2^{r_2} \cdot \dots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$. By the inductive assumption, $H \cong G_2 \times \dots \times G_{k+1}$ where $|G_i| = p_i^{r_i}$. This gives

$$G \cong G_1 \times H \cong G_1 \times G_2 \times \dots \times G_{k+1}$$

$\square$

Theorem 16.8

If G is an abelian group such that $|G| = p^n$ for some prime p then G is a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p^{k_1}} \times \mathbb{Z}_{p^{k_2}} \times \dots \times \mathbb{Z}_{p^{k_m}}$$

for some $k_1, k_2, \dots, k_m$.

Idea of Proof. Use induction with respect to the order of the group G . Let p^r be the largest order of an element in G , and let $a \in G$ be an element such that $|a| = p^r$.

One can show that there exists a short exact sequence

$$K \xrightarrow{i} G \xrightarrow{q} \mathbb{Z}_{p^r}$$

such that $q(a) = 1 \in \mathbb{Z}_{p^r}$. Take the homomorphism $s: \mathbb{Z}_{p^r} \rightarrow G$ given by $s(k) = a^k$. Since $q \circ s(k) = k$ for all $k \in \mathbb{Z}_{p^r}$, by Theorem 16.3 we obtain that $G \cong \mathbb{Z}_{p^r} \times K$. By the inductive assumption we get that K is a direct product of cyclic groups, $K \cong \mathbb{Z}_{p^{k_1}} \times \dots \times \mathbb{Z}_{p^{k_m}}$, which gives $G \cong \mathbb{Z}_{p^r} \times (\mathbb{Z}_{p^{k_1}} \times \dots \times \mathbb{Z}_{p^{k_m}})$. $\square$

Proof of Theorem 16.1. It follows from Corollary 16.7 and Theorem 16.8. $\square$

Definition 17.1

A *ring* is set R equipped with two binary operations:

- *addition*, denoted $a + b$
- *multiplication*, denoted $a \cdot b$

satisfying the following properties:

- 1) R taken with addition is an abelian group (with the identity element $0 \in R$).
- 2) Multiplication is associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for any $a, b, c \in R$.
- 3) For any $a, b, c \in R$ we have $(a + b)c = ac + bc$ and $(b + c)a = ba + ca$.

Definition 17.2

We say that a ring R is *commutative* if $ab = ba$ for any $a, b \in R$.

We say that R is a *ring with unity* if there is an element $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$.

Example. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ with the usual addition and multiplication are commutative rings with unity.

Example. $\mathbb{Z}_n$ with the addition and multiplication modulo n is a commutative ring with unity.

Example. Let $M_n(\mathbb{R})$ denote the set of $n \times n$ matrices with coefficients in $\mathbb{R}$. This is a ring with addition and multiplication given by the usual addition and multiplication of matrices. This is a ring with unity (given by the identity matrix), but it is not commutative. In the same way we can define rings $M_n(\mathbb{Z})$, $M_n(\mathbb{Q})$ and $M_n(\mathbb{C})$ of $n \times n$ matrices with integer, rational and complex coefficients.

Example. Let R be a commutative ring. By $R[x]$ we denote the ring of polynomials

with coefficients in R . Elements of $R[x]$ are polynomials of the form

$$p(x) = a_0 + a_1x + \dots + a_nx^n$$

where $a_i \in R$ and $n \geq 0$. Addition and multiplication are the usual addition and multiplication of polynomials. The ring $R[x]$ is commutative if R is commutative. If R is a ring with unit $1 \in R$ then $R[x]$ is a ring with unity given by the polynomial $p(x) = 1$.

Example. In a similar way as in the last example, from any ring R we obtain a ring $R[x_1, \dots, x_m]$ of polynomials of m variables.

Example Let S denote the set of all polynomials

$$p(x) = 0 + a_1x + a_2x^2 + \dots + a_nx^n$$

such that $a_i \in \mathbb{Z}$, $n \geq 1$. The set S is a commutative ring with the usual addition and multiplication of polynomials, but it does not have a unity.

Theorem 17.3

If a ring R has a unity, the the unity is unique.

Proof. Assume that $1, 1'$ are two unities in R , so that

$$1 \cdot a = a \cdot 1 = a \quad \text{and} \quad 1' \cdot a = a \cdot 1' = a$$

for all $a \in R$. Then $1 = 1 \cdot 1' = 1'$. □

Theorem 17.4

If a ring R . For any $a, b \in R$ we have:

- 1) $0 \cdot a = a \cdot 0 = 0$.
- 2) $a(-b) = (-a)b = -(ab)$.
- 3) $(-a)(-b) = ab$
- 4) if R has a unity $1 \in R$ then $(-1)a = a(-1) = -a$.

Proof. 1) We have

$$0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$$

Subtracting $0 \cdot a$ from both sides we obtain $0 \cdot a = 0$. By the same argument, $a \cdot 0 = 0$.

2) We have

$$a(-b) + ab = a(b - b) = a \cdot 0 = 0$$

Thus $a(-b) = -ab$. In the same way, $a(-b) = -(ab)$

3) Using part 2) we obtain $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$

4) We have

$$0 = 0 \cdot a = (1 + (-1))a = 1 \cdot a + (-1)a = a + (-1)a$$

Thus $(-1)a = -a$. Similarly, $a(-1) = -a$. □

Definition 17.5

Let R be a ring. A *subring* of R is a subset $S \subseteq R$ such that S is a ring with respect to the addition and multiplication in R .

Example. $\mathbb{Z}$ is a subring of $\mathbb{Q}$.

Example. Let $2\mathbb{Z}$ denote the set of even integers. Then $2\mathbb{Z}$ is a subring of $\mathbb{Z}$.

Theorem 17.6

Let R be a ring. A subset $S \subseteq R$ is a subring of R if and only if the following conditions are satisfied:

- 1) $0 \in S$
- 2) if $a, b \in S$ then $a + b \in S$ and $ab \in S$
- 3) if $a \in S$ then $(-a) \in S$

Definition 17.7

The *direct product* of rings R_1, R_2 is a ring $R_1 \times R_2$ defined as follows:

- Elements of $R_1 \times R_2$ are ordered tuples (a_1, a_2) where $a_i \in R_i$
- Addition and multiplication are given by

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$$

$$(a_1, a_2) \cdot (b_1, b_2) = (a_1 b_1, a_2 b_2)$$

Note. If $R_1, R_2, \dots, R_n$ are rings that the direct product $R_1 \times R_2 \times \dots R_n$ is defined analogously as as in Definiton 17.7.

Definition 18.1

Let R be a commutative ring. An element $a \neq 0$ of R is a *zero divisor* if there exists $b \neq 0$ such that $ab = 0$.

Example. In the ring $\mathbb{Z}_6$ the elements 2 and 3 are zero divisors since $2 \cdot 3 = 0$.

Definition 18.2

An *integral domain* is a commutative ring with unity which has no zero divisors.

Example. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are integral domain.

Example. $\mathbb{Z} \times \mathbb{Z}$ is not an integral domain since $(1, 0) \cdot (0, 1) = (0, 0)$.

Theorem 18.3

Let R be an integral domain and $a, b, c \in R$. If $a \neq 0$ and $ab = ac$ then $b = c$.

Proof. If $ab = ac$ then $a(b - c) = 0$. Since R is an integral domain and $a \neq 0$, this means that $b - c = 0$, and so $b = c$. $\square$

Definition 18.4

Let R be a commutative ring with unity. An element $a \in R$ is a *unit* if there exists $b \in R$ such that $ab = 1$. In such case, we denote $a^{-1} := b$.

Example The only units in $\mathbb{Z}$ are 1 and -1 .

Example Every non-zero element of $\mathbb{Q}$ is a unit.

Definition 18.5

A *field* is a commutative ring with unity in which every non-zero element is a unit.

Example. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields.

Theorem 18.6

Every field is an integral domain.

Proof. Let F be a field and $a, b \in F$ be non-zero elements. If $a \cdot b = 0$ then $b = a^{-1}(ab) = a^{-1} \cdot 0 = 0$, which is a contradiction. $\square$

Theorem 18.7

A ring $\mathbb{Z}_n$ is a field if and only if n is a prime number.

Proof. If n is not a prime number, then $n = km$ for some $1 < k, m < n$. Consider k, m as elements of $\mathbb{Z}_n$. Then $k \neq 0, m \neq 0$ but $k \cdot m = 0$. This shows that $\mathbb{Z}_n$ is not an integral domain and so it is not a field.

Conversely, assume that n is a prime number and that $k \in \mathbb{Z}_n, k \neq 0$. Then $\gcd(k, n) = 1$, so there exists $a, b \in \mathbb{Z}$ such that $ak + bn = 1$. This gives $a \cdot k = 1$ in $\mathbb{Z}_n$, so k is a unit in $\mathbb{Z}_n$ and $k^{-1} = a$. $\square$

Definition 18.8

Let F be a field with unity $1 \in F$. The *characteristic* of F is the smallest positive integer n such that

$$\underbrace{1 + 1 + \dots + 1}_{n \text{ times}} = 0$$

denote such n by $\chi(F)$.

If such n does not exist, then $\chi(F) = 0$

Example

- $\chi(\mathbb{Q}) = \chi(\mathbb{R}) = \chi(\mathbb{C}) = 0$

- If p is a prime number then $\chi(\mathbb{Z}_p) = p$.

Theorem 18.9

- 1) If F is a field then $\chi(F)$ is either 0 or a prime number.
- 2) If F is a finite field and $\chi(F) = p$ for some prime p , then F consists of p^n elements for some $n \geq 1$.

Proof. 1) Assume that $\chi(F) > 0$ and $\chi(F) = km$ for some $k, m > 1$. denote

$$\mathbf{k} = \underbrace{1 + 1 + \dots + 1}_{k \text{ times}}$$

$$\mathbf{m} = \underbrace{1 + 1 + \dots + 1}_{m \text{ times}}$$

Then $\mathbf{k}, \mathbf{m}$ are non-zero elements in F , since $k, m < \chi(F)$. Also:

$$\mathbf{km} = \underbrace{1 + 1 + \dots + 1}_{km \text{ times}} = 0$$

since $km = \chi(F)$. This means that $\mathbf{k}$ and $\mathbf{m}$ are zero divisors, which is impossible.

2) Let $|F|$ denote the number of elements in F . Let q be a prime which divides $|F|$. Consider F as an abelian group with addition. By Theorem 15.6 there exists an element $a \in F$ of order q . Notice that we have

$$pa = \underbrace{(1 + 1 + \dots + 1)}_{p \text{ times}} \cdot a = 0 \cdot a = 0$$

By Theorem 6.3 this means that q divides p . Since p and q are primes, this gives $p = q$. As a consequence $|F| = p^n$ for some $n \geq 1$. $\square$

Example. Here is an example of a field $\mathbb{Z}_3[i]$ of characteristic 3 with 9 elements. It is obtained by starting with the field $\mathbb{Z}_3$ and adding to it a new element i such that $i^2 = -1$. More explicitly:

$$\mathbb{Z}_3[i] = \{a + bi \mid a, b \in \mathbb{Z}_3\}$$

Addition and multiplication in $\mathbb{Z}_3[i]$ are given by

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i \\ (a + bi) \cdot (c + di) &= (ac - bd) + (ad + bc)i\end{aligned}$$

We will show that all non-zero elements of $\mathbb{Z}_3[i]$ are units. Notice that the polynomial $p(x) = x^2 + 1$ has no roots in $\mathbb{Z}_3$. This implies that if $a, b \in \mathbb{Z}_3$ and $b \neq 0$ then $a^2 + b^2 = b^2((ab^{-1})^2 + 1) \neq 0$. By the same argument, if $a \neq 0$ then $a^2 + b^2 \neq 0$.

Take a non-zero element $a + bi \in \mathbb{Z}_3[i]$. We have

$$(a + bi)(a - bi) = a^2 + b^2$$

Since, by the observation above, $a^2 + b^2 \neq 0$, thus the element $(a^2 + b^2)^{-1}$ exists in $\mathbb{Z}_3$ and we have

$$(a + bi)(a - bi)(a^2 + b^2)^{-1} = 1$$

This shows that $(a + bi)^{-1} = (a - bi)(a^2 + b^2)^{-1}$.

Example. The same procedure does not work for constructing a field with 25 elements. Take $\mathbb{Z}_5[i] = \{a + bi \mid a, b \in \mathbb{Z}_5\}$ with $i^2 = -1$. Then

$$(i - 2)(i - 3) = 0$$

so $\mathbb{Z}_5[i]$ is not an integral domain, and thus not a field.

The difference between $\mathbb{Z}_3[i]$ and $\mathbb{Z}_5[i]$ is that the polynomial $p(x) = x^2 + 1$ does not have any roots in $\mathbb{Z}_3$. By adding the element i to this field with $i^2 = -1$ we create roots of this polynomial, $x = i$ and $x = -i$.

On the other hand, in $\mathbb{Z}_5$ the polynomial $p(x) = x^2 + 1$ already has two roots: $x = 2$ and $x = 3$. It means that $p(x) = (x - 2)(x - 3)$. By adding i to $\mathbb{Z}_5$ we create an additional root $x = i$. This gives:

$$0 = p(i) = (i - 2)(i - 3)$$

Definition 19.1

Let R be a ring. An (*both sided*) *ideal* of R is a subring $I \subseteq R$ such that for any $a \in I$ and $r \in R$ we have $ra \in I$ and $ar \in I$.

We write $I \triangleleft R$ to denote that I is an ideal of R .

Example. For $n \geq 1$ let $n\mathbb{Z}$ denote the set of integers that are multiples of n :

$$n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$$

Then $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

In general, if R is a commutative ring and $a \in R$ then define

$$aR = \{ar \mid r \in R\}$$

Ideals of this form are called *principal ideals*.

Example. Let I, J be ideals of a ring R . Define

$$I + J = \{a + b \mid a \in I, b \in J\}$$

Then $I + J$ is an ideal of R . Notice that $I, J \subseteq I + J$.

If R is a commutative ring and $a_1, a_2, \dots, a_n \in R$ then denote:

$$\langle a_1, a_2, \dots, a_n \rangle = a_1R + a_2R + \dots + a_nR$$

We say that $\langle a_1, a_2, \dots, a_n \rangle$ is the ideal *generated by the elements* $a_1, a_2, \dots, a_n$. This ideal consists of all elements of R of the form

$$a_1r_1 + a_2r_2 + \dots + a_nr_n$$

for any $r_1, r_2, \dots, r_n \in R$. This is the smallest ideal of R containing $a_1, a_2, \dots, a_n$.

Example. The ring $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it is not an ideal of $\mathbb{Q}$.

Example. Let $R[x]$ be the ring of polynomials with coefficients on a ring R . Define

$$I = \{0 + a_1x + \dots + a_nx^n \mid a_i \in R, n \geq 0\}$$

Then I is an ideal of $R[x]$. Notice that this ideal is a principal ideal: $I = xR[x]$.

Example. In any ring R the smallest ideal is the ideal $\{0\}$, and the largest ideal is R .

Example If R is a ring with unity and $I \triangleleft R$ is an ideal such that $1 \in I$ then $I = R$. Indeed, for any $a \in R$ we have $a = a \cdot 1 \in I$.

Example. If F is a field and $I \triangleleft F$ then either $I = \{0\}$ or $I = F$. Indeed, assume that there is some $a \neq 0$ such that $a \in I$. Then for any $b \in F$ we have $b = (ba^{-1})a$ and so $b \in I$.

Note. Let R be a ring and S be a subring. Since R , taken with addition is an abelian group and S is its normal subgroup, we can consider its quotient group R/S . The elements of R/S are left cosets $a + S$ for $a \in R$ and addition is given by $(a + S) + (b + S) = (a + b) + S$.

Theorem 19.2

Let R be a rings and $I \triangleleft R$. Let $a_1, a_2, b_1, b_2 \in R$ be elements such that $a_1 + I = a_2 + I$ and $b_1 + I = b_2 + I$. Then $(a_1b_1) + I = (a_2b_2) + I$.

Note. Theorem 19.2 is not true if I is a subring of R which is not an ideal. Take for example $R = \mathbb{Q}$ and $I = \mathbb{Z}$. Let $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{2}$, $b_1 = 0$, $b_2 = 1$. We have $a_1 + \mathbb{Z} = a_2 + \mathbb{Z}$ and $b_1 + \mathbb{Z} = b_2 + \mathbb{Z}$. However, $a_1b_1 + \mathbb{Z} = 0 + \mathbb{Z}$ and $a_2b_2 = \frac{1}{2} + \mathbb{Z}$, so $a_1b_1 + \mathbb{Z} \neq a_2b_2 + \mathbb{Z}$.

Proof of Theorem 19.2. If $a_1 + I = a_2 + I$ then $a_2 = a_1 + r$ for some $r \in I$. Similarly, $b_2 = b_1 + r'$ for some $r' \in R$. This gives

$$a_2b_2 = (a_1 + r)(b_1 + r') = a_1b_1 + a_1r' + ra_2 + rr'$$

Since I is an ideal, thus $a_1r' + ra_2 + rr' \in I$, and so $a_2b_2 + I = a_1b_1 + I$. □

Definition 19.3

Let R be a ring and let $I \triangleleft R$. The *quotient ring* R/I is the ring defined as follows:

- Elements of R/I are cosets $a + I$ for $a \in R$.
- Addition: $(a + I) + (b + I) = (a + b) + I$.
- Multiplication: $(a + I) \cdot (b + I) = (ab) + I$.

Note. Let R be a ring and $I \triangleleft R$. If R is commutative then the quotient ring R/I is commutative. If R is a ring with unity $1 \in R$ then R/I is a ring with unity $(1 + I) \in R/I$.

Definition 19.4

Let R be a commutative ring with unity. A *prime ideal* of R is an ideal $I \triangleleft R$ such that $I \neq R$ and if $ab \in I$ then either $a \in I$ or $b \in I$.

Example. Take the ring of integers $\mathbb{Z}$. Recall that

$$n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$$

If p is a prime number, then $p\mathbb{Z}$ is a prime ideal. Indeed, if $ab \in p\mathbb{Z}$ then $ab = pk$ for some $k \in \mathbb{Z}$. This means that p divides ab , and so it must divide either a (in which case $a \in p\mathbb{Z}$) or b (and then $b \in p\mathbb{Z}$).

On the other if $n > 0$ is not a prime, then $n = km$ for some $0 < k, m < n$. This gives $k \notin n\mathbb{Z}$, $m \notin n\mathbb{Z}$ but $km = n \in n\mathbb{Z}$, so $n\mathbb{Z}$ is not a prime ideal.

Example. The zero ideal $\{0\} \triangleleft R$ is prime if and only if R is an integral domain. Indeed, if $\{0\}$ is not prime if and only if there are non-zero elements $a, b \in R$ such that $ab = 0$.

Theorem 19.5

If R is a commutative ring with unity and $I \triangleleft R$ then I is a prime ideal if and only if R/I is an integral domain.

Proof. Assume that R/I is not an integral domain. Then there are $a, b \in R$ such that $a + I \neq 0 + I$ and $b + I \neq 0 + I$ but $ab + I = 0 + I$. This means that $a \notin I$, $b \notin I$ but $ab \in I$, which shows that I is not a prime ideal.

The prove of the converse is similar. □

Definition 19.6

Let R be a commutative ring with unity. A *maximal ideal* of R is an ideal $I \triangleleft R$ such that $I \neq R$ if $J \triangleleft R$ is any other ideal such that $J \neq R$ and $I \subseteq J$, then $I = J$.

Example. The ideal $4\mathbb{Z} \triangleleft \mathbb{Z}$ is not maximal since $4\mathbb{Z} \subseteq 2\mathbb{Z}$.

Example. Take the ring $\mathbb{Z}$. We will show that if p is a prime then $p\mathbb{Z}$ is a maximal ideal. Indeed, assume that there is some ideal $J \triangleleft \mathbb{Z}$ such that $p\mathbb{Z} \subseteq J$. Let $a \in J$ and $a \notin p\mathbb{Z}$. Then a is not divisible by p , so $\gcd(p, a) = 1$. As a consequence we can find $k, l \in \mathbb{Z}$ such that $pk + al = 1$. Since $p, a \in J$, this gives $1 \in J$, and so $J = \mathbb{Z}$.

Theorem 19.7

If R is a commutative ring with unity and $I \triangleleft R$ then I is a maximal ideal if and only if R/I is a field.

Proof. Assume that R/I is a field and let $J \triangleleft R$ be an ideal such that $I \subseteq J$ but $I \neq J$. We will show that $J = R$. Indeed, take $a \in J$ such that $a \notin I$. Then $a + I$ is a non-zero element in R/I , so it is a unit in R/I . In other words, there is $b \in R$ such that $(ab + I) = (a + I)(b + I) = 1 + I$. This means that $1 = ab + c$ for some $c \in I$. Since $a \in J$ and $c \in J$ we obtain that $1 \in J$, and so $J = R$.

Conversely, assume that I is a maximal ideal and let $a + I \in R/I$ be a non-zero element. Take $J = I + aR$. Then $I \subseteq J$ and $I \neq J$ since $a \notin I$. By maximality of I we obtain that $J = R$. This means that $1 \in J$, so $1 = c + ab$ for some $c \in I$ and $b \in R$. This gives:

$$1 + I = ab + I = (a + I)(b + I)$$

so $a + I$ is a unit in R/I with $(a + I)^{-1} = b + I$. □

Theorem 19.8

If R is a commutative ring with unity then every maximal ideal of R is a prime ideal.

Proof. If $I \triangleleft R$ is a maximal ideal then R/I is a field by Theorem 19.7, and so by Theorem 18.6 it is an integral domain. Then, Theorem 19.5 implies that I is a prime ideal. □

Example. In the ring $\mathbb{Z}[x]$ define

$$I = \{p(x) \in \mathbb{Z}[x] \mid p(0) = 0\}$$

This is an ideal of $\mathbb{Z}[x]$. We claim that I is a prime ideal. Indeed, if $p(x), q(x) \in I$ are polynomials such that $p(x)q(x) \in I$, then $p(0) \cdot q(0) = 0$, so either $p(0) = 0$ or $q(0) = 0$. This means that either $p(x) \in I$ or $q(x) \in I$.

On the other hand, I is not a maximal ideal. Take, for example

$$J = \{p(x) \in \mathbb{Z}[x] \mid p(0) \text{ is an even number}\}$$

One can check that J is an ideal of $\mathbb{Z}[x]$. We have $I \subseteq J$, but $J \neq \mathbb{Z}[x]$, since e.g. $p(x) = 1 + x \notin J$.

Definition 20.1

A *homomorphism* from a ring R to a ring S is a function $f: R \rightarrow S$ such that for any $a, b \in R$ we have

- $f(a + b) = f(a) + f(b)$
- $f(ab) = f(a) \cdot f(b)$

Note. Since a homomorphism of rings $f: R \rightarrow S$ is a homomorphism of their additive groups, thus we have $f(0) = 0$ and $f(-a) = -f(a)$ for any $a \in R$.

On the other hand if R and S are rings with unity, then it need not be true in general that $f(1) = 1$. Take for example $R = \mathbb{Z}$, $S = \mathbb{Z} \times \mathbb{Z}$, and let $f: \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ be given by $f(n) = (n, 0)$. Then f is a ring homomorphism, but $f(1) = (1, 0)$ which is not the unity in $\mathbb{Z} \times \mathbb{Z}$.

To avoid such situations, usually, when working with rings with unity, it is additionally assumed that homomorphisms preserve the unity, $f(1) = 1$.

Example. For $n > 1$ the function $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $f(k) = k \bmod n$ is a ring homomorphism.

Example. Let R be a ring and let $a \in R$. The function $f: R[x] \rightarrow R$ defined by $f(p(x)) = p(a)$ is a homomorphism of rings.

Example. If R is a ring and $I \triangleright R$ then the function $q: R \rightarrow R/I$ given by $q(a) = a + I$ is a homomorphism of rings.

Definition 20.2

A *isomorphism* of rings is a homomorphism $f: R \rightarrow S$ which is a bijection.

If there exists an isomorphism between rings R and S then we say that these rings are *isomorphic* and we write $R \cong S$.

Theorem 20.3

If $f: R \rightarrow S$ is an isomorphism of rings then the inverse function $f^{-1}: S \rightarrow R$ is also an isomorphism of rings.

Proof. Exercise. □

Definition 20.4

Let $f: R \rightarrow S$ be a homomorphism of rings. The *image* of f is the set $\text{Im}(f) \subseteq S$ defined by

$$\text{Im}(f) = \{f(r) \mid r \in R\}$$

The *kernel* of f is the set $\text{Ker}(f) \subseteq R$ given by

$$\text{Ker}(f) = \{r \in R \mid f(r) = 0\}$$

Theorem 20.5

Let $f: R \rightarrow S$ be homomorphism of rings. Then

- 1) $\text{Im}(f)$ is a subring of S
- 2) $\text{Ker}(f)$ is an ideal of R .

Proof.

1) Exercise.

2) One can check that $\text{Ker}(f)$ is a subring of R (exercise). Also, if $a \in \text{Ker}(f)$ and $r \in R$ then

$$f(ra) = f(r) \cdot f(a) = f(r) \cdot 0 = 0$$

so $ra \in \text{Ker}(f)$. Similarly, $ar \in \text{Ker}(f)$. This show that $\text{Ker}(f) \triangleleft R$. □

Theorem 20.6 (First Isomorphism Theorem for Rings)

Let $f: R \rightarrow S$ be a ring which is onto. Then $S \cong R/\text{Ker}(f)$.

Proof. Define $g: R/\text{Ker}(f) \rightarrow S$ by $g(a + \text{Ker}(f)) = f(a)$. One can check that this is a well defined function, which gives an isomorphism of rings. □

Example. Recall that for $n > 1$ we have an onto homomorphism $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$, $f(k) = k \bmod n$. Notice that

$$\text{Ker}(f) = \{nk \mid k \in \mathbb{Z}\} = n\mathbb{Z}$$

This gives $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Example. Take the homomorphism $f: R[x] \rightarrow R$ defined by $f(p(x)) = p(0)$. This homomorphism is onto, since if $a \in R$ then $r = f(p(x))$ for the polynomial $p(x) = a$. We have

$$\begin{aligned} \text{Ker}(f) &= \{p(x) \mid p(x) = 0\} \\ &= \{a_1x + \dots + a_nx^n \mid a_i \in R, n \geq 0\} \\ &= xR[x] \end{aligned}$$

This shows that $R[x]/xR[x] \cong R$.

Theorem 20.7

If R is a ring and $I \triangleleft R$ then there exists a ring homomorphism $f: R \rightarrow S$ such that $\text{Ker}(f) = I$.

Proof. Take $S = R/I$ and $f: R \rightarrow R/I$ defined by $f(a) = a + I$. □

Definition 21.1

Let R be a commutative ring. The *ring of polynomials* $R[x]$ of variable x with coefficient in R , defined as follows.

- Elements of $R[x]$ are expressions of the form

$$p(x) = a_n x^n + a_{n-1} x_{n-1} + \dots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i$$

where $n \geq 0$.

- Addition in $R[x]$: if $p(x) = \sum_{i=0}^n a_i x^i$, $q(x) = \sum_{i=0}^m b_i x^i$ then

$$p(x) + q(x) = \sum_{i=0}^s (a_i + b_i) x^i$$

where $s = \max(m, n)$. In this formula, if $i > n$ then we take $a_i = 0$ and if $i > m$ then we take $b_i = 0$.

- Multiplication in $R[x]$: if $p(x) = \sum_{i=0}^n a_i x^i$, $q(x) = \sum_{i=0}^m b_i x^i$ then

$$p(x) \cdot q(x) = \sum_{i=0}^s c_i x^i$$

where $s = m + n$ and $c_i = a_0 b_i + a_1 b_{i-1} + \dots + a_i b_0$

Note. Let R be a commutative ring. Any polynomial $p(x) = a_n x^n + a_{n-1} x_{n-1} + \dots + a_0$ defines a function $\bar{p}: R \rightarrow R$ given by $\bar{p}(r) = a_n r^n + a_{n-1} r^{n-1} + \dots + a_0$. In general it may happen that $p(x)$, $q(x)$ are different polynomials, but the functions $\bar{p}$, $\bar{q}$ they define are the same.

For example, take $p(x) = 0$, $q(x) = x^2 + x \in \mathbb{Z}_2[x]$. Then $p(x) \neq q(x)$. On the other hand, consider the functions $\bar{p}, \bar{q}: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$. We have $\bar{p}(0) = 0 = \bar{q}(0)$ and $\bar{p}(1) = 0 = \bar{q}(1)$, so $\bar{p} = \bar{q}$.

Definition 21.2

For a polynomial $p(x) = \sum_i a_i x^i \in R[x]$ such that $p(x) \neq 0$, the *degree* $p(x)$ is the integer $n \geq 0$ such $a_n \neq 0$ and $a_i = 0$ for all $i > n$. We denote $\deg p(x) = n$. For the zero polynomial $p(x) = 0$ degree is not defined.

Theorem 21.3

Let R be an integral domain and let $p(x), q(x) \in R[x]$ be non-zero polynomials. Then

$$\deg(p(x) \cdot q(x)) = \deg p(x) + \deg q(x)$$

Proof. If $\deg p(x) = n$ and $\deg q(x) = m$ then $p(x) = a_n x^n + \dots + a_0$ $q(x) = b_m x^m + \dots + b_0$ for some $a_i, b_i \in R$ such that $a_n \neq 0, b_m \neq 0$. Then

$$p(x) \cdot q(x) = a_n b_m x^{m+n} + \dots + a_0 b_0$$

Since R is an integral domain, thus $a_n b_m \neq 0$, so $\deg(p(x) \cdot q(x)) = m + n$. $\square$

Example. Theorem 21.3 is not true when R is not an integral domain. Take e.g. $p(x) = 2x + 1, q(x) = 3x + 1 \in \mathbb{Z}_6[x]$. Then $\deg p(x) = \deg q(x) = 1$ and $\deg(p(x) \cdot q(x)) = \deg(5x + 1) = 1$.

Corollary 21.4

If R is an integral domain then $R[x]$ is also an integral domain.

Proof. If $p(x), q(x) \in R[x]$ are non-zero polynomials, then by Theorem 21.3 $\deg(p(x) \cdot q(x))$ is defined, so $p(x) \cdot q(x) \neq 0$. $\square$

Theorem 21.5

Let R be an integral domain. Let $p(x) = a_n x^n + \dots + a_0$ and be a polynomial in $R[x]$ such that $p(x) \neq 0$. and a_n is a unit. Then for any $g(x) \in R[x]$ there exist unique polynomials $q(x), r(x) \in R[x]$ such that

$$g(x) = q(x)p(x) + r(x)$$

where either $r(x) = 0$ or $\deg r(x) < \deg p(x)$.

Note. We say that $q(x)$ is the *quotient* and $r(x)$ is the *remainder* of the division of $g(x)$ by $p(x)$.

Proof of Theorem 21.5. To show existence of $q(x)$ and $r(x)$, we argue by induction with respect to $\deg g(x)$. If $g(x) = 0$ or $\deg g(x) < \deg p(x)$ then we take $q(x) = 0$ and $r(x) = g(x)$. Next, assume that for any polynomial $g'(x)$ of degree smaller than m we can find $q(x)$ and $r(x)$ as in the theorem, and let $g(x) = b_mx^m + \dots + b_0$ be a polynomial such that $\deg g(x) = m \geq n = \deg p(x)$. Let

$$g(x) = b_mx^m + \dots + b_0$$

Using the assumption that a_n is a unit in R , take the polynomial

$$s(x) = g(x) - (b_ma_n^{-1})p(x) \cdot x^{m-n}$$

We have:

$$\begin{aligned} s(x) &= g(x) - (b_ma_n^{-1})p(x) \cdot x^{m-n} \\ &= (b_mx^m + b_{m-1}x^{m-1} + \dots) - (b_ma_n^{-1})(a_nx^n + a_{n-1}x^{n-1} + \dots) \cdot x^{m-n} \\ &= (b_mx^m + b_{m-1}x^{m-1} + \dots) - ((b_ma_n^{-1}a_n)x^m + (b_ma_n^{-1}a_{n-1})x^{m-1} + \dots) \\ &= (b_mx^m + b_{m-1}x^{m-1} + \dots) - (b_mx^m + (b_ma_n^{-1}a_{n-1})x^{m-1} + \dots) \\ &= ((b_{m-1} - b_ma_n^{-1}a_{n-1})x^{m-1} + \dots) \end{aligned}$$

This shows that $\deg s(x) \leq m$ and so, by the inductive assumption, we have $s(x) = q'(x)p(x) + r'(x)$ for some $q'(x), r'(x) \in R[x]$ such that either $r'(x) = 0$ or $\deg r'(x) < \deg p(x)$. This gives:

$$\begin{aligned} g(x) &= s(x) + (b_ma_n^{-1})p(x) \cdot x^{m-n} \\ &= (q'(x)p(x) + r'(x)) + (b_ma_n^{-1})p(x) \cdot x^{m-n} \\ &= (q'(x) + (b_ma_n^{-1})x^{m-n})p(x) + r'(x) \end{aligned}$$

Thus we can take $q(x) = q'(x) + (b_ma_n^{-1})x^{m-n}$ and $r(x) = r'(x)$.

For uniqueness, assume that $g(x) = q_1(x)p(x) + r_1(x)$ and $g(x) = q_2(x)p(x) + r_2(x)$ for some $p_1(x), p_2(x), r_1(x), r_2(x) \in R[x]$. Then

$$0 = (q_1(x) - q_2(x))p(x) + (r_1(x) - r_2(x))$$

or equivalently

$$(q_1(x) - q_2(x))p(x) = -(r_1(x) - r_2(x))$$

If $q_1(x) \neq q_2(x)$ then degree of the left hand side is greater or equal to $\deg p(x)$, which is greater than the degree of the right hand side. Since this is impossible, we get $q_1(x) = q_2(x)$. This implies that $r_1(x) = r_2(x)$. $\square$

Exercise. Let $p(x) = x^2 + 3x + 2$ and $g(x) = 3x^4 + 2x^2 - x + 7$ be polynomials in $\mathbb{Z}[x]$. Find the quotient and the remainder of the division of $g(x)$ by $p(x)$.

Exercise. Let $p(x) = 4x^2 + 3x + 2$ and $g(x) = 3x^4 + 2x^2 + 4x + 1$ be polynomials in $\mathbb{Z}_5[x]$. Find the quotient and the remainder of the division of $g(x)$ by $p(x)$.

Definition 21.6

Let R be an integral domain and let $p(x), g(x) \in R[x]$. We say that $p(x)$ *divides* $g(x)$ if there is $q(x) \in R[x]$ such that $g(x) = q(x)p(x)$.

Definition 21.7

Let R be an integral domain and let $p(x) \in R[x]$. We say that an element $a \in R$ is a *root* of $p(x)$ if $p(a) = 0$.

Theorem 21.8

Let R be an integral domain and let $p(x) \in R[x]$. An element $a \in R$ is a root of $p(x)$ if and only if $(x - a)$ divides $p(x)$.

Proof. By Theorem 21.5 we have

$$p(x) = q(x)(x - a) + r(x)$$

for some $q(x), r(x) \in R[x]$ where either $r(x) = b$ for some $b \in R$. This gives:

$$p(a) = q(a)(a - a) + b = b$$

Thus $p(a) = 0$ if and only if $b = 0$. In such case $p(x) = q(x)(x - a)$. □

Corollary 21.9

Let R be an integral domain, let $p(x) \in R[x]$ and let $a_1, \dots, a_m \in R$ be distinct elements of R . Then $a_1, \dots, a_m$ are roots of $p(x)$ if and only if $(x - a_1) \cdots (x - a_m)$ divides $p(x)$.

Proof. If $(x - a_1) \cdots (x - a_m)$ divides $p(x)$ then

$$p(x) = q(x) \cdot (x - a_1) \cdots (x - a_m)$$

for some $q(x) \in R[x]$. Then $p(a_i) = 0$ for $i = 1, \dots, m$, so $a_1, \dots, a_m$ are roots of $p(x)$. Conversely, assume that $a_1, \dots, a_m$ are roots of $p(x)$. By Theorem 21.8 we have

$$p(x) = q_1(x) \cdot (x - a_1)$$

for some $q_1 \in R[x]$. This gives:

$$0 = p(a_2) = q_1(a_2) \cdot (a_2 - a_1)$$

Since $a_1 \neq a_2$, we have $a_2 - a_1 \neq 0$. Thus, since R is an integral domain, we obtain that $q_1(a_2) = 0$. Applying Theorem 21.8 to the polynomial $q_1(x)$ we obtain that

$$q_1(x) = q_2(x) \cdot (x - a_2)$$

for some $q_2(x) \in R[x]$, and so

$$p(x) = q_2(x) \cdot (x - a_2) \cdot (x - a_1)$$

Continuing this argument inductively, we obtain that

$$p(x) = q(x) \cdot (x - a_1) \cdot \dots \cdot (x - a_m)$$

for some $q(x) \in R[x]$. □

Corollary 21.10

If R is an integral domain and $p(x) \in R[x]$ is a non-zero polynomial, then $p(x)$ has at most $\deg p(x)$ distinct roots.

Proof. If $a_1, \dots, a_m$ are distinct roots of $p(x)$ then by Corollary 21.9 we have

$$p(x) = q(x) \cdot (x - a_1) \cdot \dots \cdot (x - a_m)$$

Then $\deg p(x) = \deg q(x) + m$, so $m \leq \deg p(x)$. □

Corollary 21.11

Let R be an integral domain consisting of infinitely many elements. If $p(x), g(x) \in R[x]$ are polynomials such that $p(a) = g(a)$ for all $a \in R$ then $p(x) = g(x)$.

Proof. Assume that $p(a) = g(a)$ for all $a \in R$. Take $f(x) = p(x) - g(x)$. Then $f(a) = 0$ for all $a \in R$. Since R consists of infinitely many elements, thus $f(a)$ has infinitely many roots. By 21.10 this is possible only if $f(x) = 0$, i.e. $p(x) = g(x)$. $\square$

Recall that if R a commutative ring, then an ideal $J \triangleleft R$ is principal if J is generated by a single element. That is, there is $a \in R$ such that

$$J = \langle a \rangle = \{ar \mid r \in R\}$$

Definition 21.12

A ring R is a *principal ideal domain (PID)* if R is an integral domain and every ideal of R is principal.

Example. Every field is a PID. Indeed, if F is a field then the only ideals of F are $\{0\} = \langle 0 \rangle$ and $F = \langle 1 \rangle$.

Example. The ring of integers $\mathbb{Z}$ is a PID. Indeed, let $J \triangleleft \mathbb{Z}$. If $J = \{0\}$ then $J = \langle 0 \rangle$. If $J \neq \{0\}$, let a be the smallest positive integer such that $a \in J$. We will show that $J = \langle a \rangle$. Indeed, assume that $b \in J$. We have $b = qa + r$ for some $q, r \in \mathbb{Z}$ such that $0 \leq r < a$. Since $r = b - qa$, thus $r \in J$. Since a is the smallest positive element of J , we must have $r = 0$. Thus $b = qa$, and so $b \in \langle a \rangle$. This shows that $J \subseteq \langle a \rangle$. Also, since $a \in J$, thus $\langle a \rangle \subseteq J$. This gives $J = \langle a \rangle$.

Example. The ring $\mathbb{Z}[x]$ is not a PID. Take for example the ideal $\langle 2, x \rangle \triangleleft \mathbb{Z}[x]$ generated by the constant polynomial $p(x) = 2$ and the polynomial $q(x) = x$. Elements of $\langle 2, x \rangle$ are polynomials $g(x) = a_n x^n + \dots + a_0$ such that a_0 is an even number. The ideal $\langle 2, x \rangle$ is not principal. Indeed, assume that $\langle 2, x \rangle = \langle f(x) \rangle$ for some $f(x) \in \mathbb{Z}[x]$. Since $2 \in \langle 2, x \rangle$, thus $f(x)$ must divide 2. This means that $f(x)$ is a constant polynomial, and $f(x) = 1, -1, 2$ or -2 . If $f(x) = \pm 1$ then $\langle f(x) \rangle = \mathbb{Z}[x] \neq \langle 2, x \rangle$. Also, if $f(x) = \pm 2$, then $\langle f(x) \rangle$ consists of polynomials whose all coefficients are even. Thus $\langle f(x) \rangle \neq \langle 2, x \rangle$.

Theorem 21.13

If F is a field then $F[x]$ is a PID.

Note. Theorem 21.13 can be considered as a generalization of Theorem 21.8 as follows. For $a \in F$ consider the homomorphism $\varphi: F[x] \rightarrow F$ given by $\varphi(p(x)) = p(a)$. Then $\text{Ker}(\varphi)$ is an ideal of $F[x]$ consisting of polynomials $g(x)$ such that $g(a) = 0$.

Theorem 21.8 says that every such polynomial is a multiple of $(x - a)$, so $\text{Ker}(\varphi)$ is a principal ideal, $\text{Ker}(\varphi) = \langle (x - a) \rangle$. Now, if $J \triangleleft F[x]$ is an arbitrary ideal, then there is a homomorphism $\varphi: F[x] \rightarrow S$ for some ring S such that $J = \text{Ker}(\varphi)$. Theorem 21.13 says that $\text{Ker}(\varphi) = \langle p(x) \rangle$ for some $p(x) \in F[x]$.

Proof of Theorem 21.13. Let $J \triangleleft F[x]$. If $J = \{0\}$ then $J = \langle 0 \rangle$. Assume then that $J \neq \{0\}$. Let $p(x)$ be a non-zero polynomial of the smallest degree, such that $p(x) \in J$. We will show that $J = \langle p(x) \rangle$.

Let $f(x) \in J$. We need to show that $f(x) = q(x)p(x)$ for some $q(x) \in F[x]$. Since F is a field, the coefficient of the highest degree term of $p(x)$ is a unit, so by Theorem 21.5 we have $f(x) = q(x)p(x) + r(x)$ where either $r(x) = 0$ or $\deg r(x) < \deg p(x)$. Assume that $r(x) \neq 0$. Then $r(x) = f(x) - q(x)p(x) \in J$, which is impossible, since by assumption $p(x)$ is a polynomial of the smallest degree in J . Thus $r(x) = 0$ and so $f(x) = q(x)p(x)$. $\square$

Definition 22.1

Let R be an integral domain. An element $a \in R$ is *irreducible* if $a \neq 0$, a is not a unit and if $a = bc$ for some $b, c \in R$ then either b or c is a unit.

Example.

- $n \in \mathbb{Z}$ is irreducible and only if $n = \pm p$ where p is a prime number.
- A field has no irreducible elements.
- Take $p(x) \in \mathbb{R}[x]$, $p(x) = x^2 + 1$. Then $p(x)$ is irreducible in $\mathbb{R}[x]$.
- Take $p(x) \in \mathbb{C}[x]$, $p(x) = x^2 + 1$. Then $p(x)$ is not irreducible in $\mathbb{C}[x]$: since $p(x) = (x - i)(x + i)$.

Theorem 22.2

If R is an integral domain, $a \in R$ is irreducible and $u \in R$ is a unit then ua is irreducible.

Proof. Let $ua = bc$. We need to show that either b or c is a unit. Since $a = (u^{-1}b)c$ and a is irreducible, thus either c is a unit or $u^{-1}b$ is a unit. In the second case, since a product of units is a unit, we obtain that $b = u(u^{-1}b)$ is a unit. $\square$

Definition 22.3

Let R be an integral domain. Elements $a, b \in R$ are *associates* if $a = ub$ for some unit $u \in R$. We write: $a \sim b$.

Example.

- If $m, n \in \mathbb{Z}$ then $m \sim n$ iff $m = \pm n$.
- Check: units in $\mathbb{R}[x]$ are non-zero polynomials of degree 0. It follows that if $p(x), q(x) \in \mathbb{R}[x]$ then $p(x) \sim q(x)$ iff $p(x) = aq(x)$ for some $a \in \mathbb{R} - \{0\}$.

Definition 22.4

A *unique factorization domain (UFD)* is an integral domain R that satisfies the following conditions:

1) if $a \in R$ is a non-zero, non-unit element then

$$a = b_1 \cdot \dots \cdot b_k$$

for some irreducible elements $b_1, \dots, b_k \in R$

2) if $b_1, \dots, b_k, c_1, \dots, c_l$ are irreducible elements such that

$$b_1 \cdot \dots \cdot b_k = c_1 \cdot \dots \cdot c_l$$

then $k = l$ and for some permutation $\sigma: \{1, \dots, k\} \rightarrow \{1, \dots, k\}$ we have $b_1 \sim c_{\sigma(1)}, \dots, b_k \sim c_{\sigma(k)}$.

Example.

- $\mathbb{Z}$ is a UFD by the Fundamental Theorem of Arithmetic.
- If F is a field then F is a UFD since all non-zero elements of F are units.

Note. For an positive integer K let $\mathbb{Z}[\sqrt{-K}]$ denote the ring defined as follow:

- Elements of $\mathbb{Z}[\sqrt{-d}]$ are expressions $a + b\sqrt{K}i$ for $a, b \in \mathbb{Z}$.
- Addition:

$$(a + b\sqrt{n}i) + (c + d\sqrt{K}i) = (a + c) + (b + d)\sqrt{K}i$$

- Multiplication:

$$(a + b\sqrt{n}i) \cdot (c + d\sqrt{K}i) = (ac - nbd) + (ad + bc)\sqrt{K}i$$

The ring $\mathbb{Z}[\sqrt{-K}]$ is a subring of the field of complex numbers $\mathbb{C}$, so it is an integral domain.

Theorem 22.5

The ring $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Proof. For $a + b\sqrt{5}i \in \mathbb{Z}[\sqrt{-5}]$ define

$$N(a + b\sqrt{5}i) = (a + b\sqrt{5}i)(a - b\sqrt{5}i) = a^2 + 5b^2 \in \mathbb{N}$$

Notice that for any $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$ we have:

- 1) $N(\alpha) = 1$ iff $\alpha = \pm 1$
- 2) $N(\alpha) = 0$ iff $\alpha = 0$
- 3) $N(\alpha\beta) = N(\alpha)N(\beta)$
- 4) $N(\alpha) \neq 3$ for all $\alpha \in \mathbb{Z}[\sqrt{-5}]$

Observation 1. The only units in $\mathbb{Z}[\sqrt{-5}]$ are 1 and -1 .

Indeed, if $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is a unit then

$$N(\alpha)N(\alpha^{-1}) = N(\alpha\alpha^{-1}) = N(1) = 1$$

Therefore $N(\alpha) = 1$, and so $\alpha = \pm 1$.

As a consequence we obtain that in $\mathbb{Z}[\sqrt{-5}]$ we have $\alpha \sim \beta$ if and only if $\alpha = \pm\beta$.

Observation 2. If $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is an element such that $N(\alpha) = 9$ then α is irreducible.

Indeed, if $\alpha = \beta\beta'$ then

$$N(\beta)N(\beta') = N(\alpha) = 9$$

Therefore $N(\beta)$ must be either 1 (and so β is a unit), 3 (impossible), or 9 (and then $N(\beta') = 1$, i.e. β' is a unit).

Take $9 \in \mathbb{Z}[\sqrt{-5}]$. We have

$$3 \cdot 3 = 9 = (2 + \sqrt{5}i)(2 - \sqrt{5}i)$$

By Observation 2 the elements 3, $2 + \sqrt{5}i$, $2 - \sqrt{5}i$ are irreducible in $\mathbb{Z}[\sqrt{-5}]$. On the other hand, by Observation 1 we obtain

$$3 \not\sim (2 + \sqrt{5}i), \quad 3 \not\sim (2 - \sqrt{5}i)$$

As a consequence 9 does not have a unique factorization in $\mathbb{Z}[\sqrt{-5}]$. □

Definition 23.1

Let R be an integral domain, and let $a, b \in R$. We say that a divides b if $b = ac$ for some $c \in R$. We then write: $a \mid b$.

Theorem 23.2

If R is an integral domain and $a, b \in R$ are non-zero elements then $a \sim b$ and only if $a \mid b$ and $b \mid a$.

Proof. If $a \mid b$ and $b \mid a$ then $b = ca$ and $a = db$. This gives $b = cdb$. Since R is an integral domain, we obtain that $cd = 1$, so c, d are units and $d = c^{-1}$. Therefore $a \sim b$.

Conversely, if $a \sim b$ then $b = ua$ for some unit u . Then $a \mid b$. Also, $a = u^{-1}b$, so $b \mid a$. $\square$

Example

- In $\mathbb{Z}$ we have:

$$\{\text{prime elements}\} = \{\pm \text{prime numbers}\} = \{\text{irreducible elements}\}$$

- By the proof of Theorem 22.5, in $\mathbb{Z}[\sqrt{-5}]$ the element $\alpha = 2 + \sqrt{5}i$ is irreducible. On the other hand α is not a prime element since $\alpha \mid (3 \cdot 3)$ but $\alpha \nmid 3$.

Theorem 23.3

If R is an integral domain and $a \in R$ is a prime element then a is irreducible.

Proof. Let $a \in R$ be a prime element and let $a = bc$. We want to show that either b or c must be a unit in R .

We have $a \mid (bc)$. Since a is a prime element it implies that $a \mid b$ or $a \mid c$.

We can assume that $a \mid b$. Since also $b \mid a$, thus by Theorem 23.2 we obtain that $a \sim b$, i.e. $a = bu$ for some unit $u \in R$. Therefore we have

$$bc = a = bu$$

Since R is an integral domain, this gives $u = c$, and so c is a unit. □

Theorem 23.4

If R is a UFD and $a \in R$ then a is an irreducible element if and only if a is a prime element.

Proof. ($\Leftarrow$) This follows from Theorem 23.3.

($\Rightarrow$) Assume that $a \in R$ is irreducible and that $a \mid (bc)$. We want to show that either $a \mid b$ or $a \mid c$.

If $b = 0$, then $b = a \cdot 0$ so $a \mid b$. If b is a unit, then $c = b^{-1}bc$ so $a \mid c$.

As a consequence we can assume that b, c are non-zero, non-units.

Since $a \mid (bc)$ there is $d \in R$ such that $bc = ad$. Assume that d is not a unit. Since R is a UFD we have decompositions:

$$b = b_1 \cdot \dots \cdot b_m, \quad c = c_1 \cdot \dots \cdot c_n, \quad d = d_1 \cdot \dots \cdot d_p$$

where b_i, c_j, d_k are irreducible. This gives

$$b_1 \cdot \dots \cdot b_m \cdot c_1 \cdot \dots \cdot c_n = a \cdot d_1 \cdot \dots \cdot d_p$$

By the uniqueness of decomposition in UFDs this implies that either $a \sim b_i$ for some i or $a \sim c_j$ for some j . In the first case we get $a \mid b$, and in the second case $a \mid c$.

If d is a unit the argument is similar. □

Theorem 23.5

An integral domain R is a UFD if and only if the following conditions are satisfied:

- 1) every non-zero, non-unit element of R is a product of irreducible elements.
- 2) every irreducible element in R is a prime element.

Proof. ($\Rightarrow$) This follows from the definition of UFD and Theorem 23.4.

($\Leftarrow$) Assume that R satisfies conditions 1) - 2) of the theorem. We only need to show that if $b_1, \dots, b_k, c_1, \dots, c_l$ are irreducible elements in R such that

$$b_1 \cdot \dots \cdot b_k = c_1 \cdot \dots \cdot c_l$$

then $k = l$, and after reordering of the factors we have $b_1 \sim c_1, \dots, b_k \sim c_k$.

We argue by induction with respect to k .

If $k = 1$ then we have $b_1 = c_1 \cdot \dots \cdot c_l$. Since b_1 is irreducible, this implies that $l = 1$, and so $b_1 = c_1$.

Next, assume that the uniqueness property holds for some k and that we have

$$b_1 \cdot \dots \cdot b_k \cdot b_{k+1} = c_1 \cdot \dots \cdot c_l$$

where b_i, c_j are irreducible elements. This implies that $b_{k+1} \mid (c_1 \cdot \dots \cdot c_l)$. By condition 2) we get that b_{k+1} is a prime element. It follows that $b_{k+1} \mid c_j$ for some $1 \leq j \leq l$. We can assume that $b_{k+1} \mid c_l$. Then $c_l = ab_{k+1}$ for some $a \in R$. Since c_l, b_{k+1} are irreducible, a must be a unit. This shows that $b_{k+1} \sim c_l$. Furthermore, we obtain from here that

$$b_1 \cdot \dots \cdot b_k \cdot b_{k+1} = c_1 \cdot \dots \cdot c_{l-1} \cdot ab_{k+1}$$

Since R is an integral domain this gives

$$b_1 \cdot \dots \cdot b_k = c_1 \cdot \dots \cdot c_{l-1}a$$

Since b_k is irreducible and a is a unit, the product $c_{l-1}a$ is an irreducible element. Therefore, by the inductive assumption we get that $k = l - 1$, and that after reordering of factors we have

$$b_1 \sim c_1, \dots, b_{k-1} \sim c_{k-1}, b_k \sim c_{l-1}a \sim c_{l-1}$$

□

Recall: A principal ideal domain (PID) is a ring R which is an integral domain, such that every ideal $I \triangleleft R$ is principal, i.e. $I = \langle a \rangle$ for some $a \in R$.

Theorem 24.1

If R is a PID then it is a UFD.

Lemma 24.2

If R is a PID and $I_1, I_2, \dots$ are ideals of R such that

$$I_1 \subseteq I_2 \subseteq \dots$$

then there exists $n \geq 1$ such that $I_n = I_{n+1} = \dots$

Proof. Take $J = \bigcup_{i=1}^{\infty} I_i$. One can check that J is an ideal of R . Since R is a PID we have $J = \langle a \rangle$ for some $a \in I$. Take $n \geq 1$ such that $a \in I_n$. Then we get

$$J \subseteq I_n \subseteq I_{n+1} \subseteq \dots \subseteq J$$

It follows that $I_n = I_{n+1} = \dots = J$. □

Lemma 24.3

Let R be a PID. An element $a \in R$ is irreducible if and only if $\langle a \rangle$ is a maximal ideal of R .

Proof. Exercise. □

Proof of Theorem 24.1. Let R be a PID. By Theorem 23.5 it suffices to show that

- 1) Every non-zero, non-unit element of R is a product of irreducible elements
- 2) Every irreducible element in R is a prime element.

1) We argue by contradiction. Assume that $a_0 \in R$ is a non-zero, non-unit element that is not a product of irreducibles. This implies that $a_0 = a_1 b_1$ for some non-zero, non-unit elements $a_1, b_1 \in R$.

If both a_1 and b_1 were products of irreducibles, then a_0 would be also a product of irreducibles, contradicting our assumption. We can then assume that a_1 is not a product of irreducibles, and so in particular we have $a_1 = a_2 b_2$ for some non-zero, non-unit elements $a_2, b_2 \in R$.

By induction we obtain that for $i = 1, 2, \dots$ there exists non-zero, non-unit elements $a_i, b_i \in R$ such that $a_i = a_{i+1} b_{i+1}$ for all $i \geq 0$.

Consider the chain of ideals

$$\langle a_0 \rangle \subseteq \langle a_1 \rangle \subseteq \dots$$

By Lemma 24.3 we obtain that $\langle a_n \rangle = \langle a_{n+1} \rangle$ for some $n \geq 0$. This means that $a_n = a_{n+1} u$ for some unit $u \in R$ (check!). As a consequence $a_{n+1} b_{n+1} = a_n = a_{n+1} u$ and so $b_{n+1} = u$. This is a contradiction, since b_{n+1} is not a unit.

2) Let $a \in R$ be an irreducible element and let $a \mid (bc)$. We need to show that either $a \mid b$ or $a \mid c$.

Assume that $a \nmid b$. This implies that $b \notin \langle a \rangle$ and so $\langle a \rangle \neq \langle a \rangle + \langle b \rangle$

Since by Lemma 24.3 the ideal $\langle a \rangle$ is a maximal ideal, we obtain then that $\langle a \rangle + \langle b \rangle = R$, and so in particular $1 \in \langle a \rangle + \langle b \rangle$. Therefore $1 = ar + bs$ for some $r, s \in R$, and so $c = a(rc) + (bc)s$. Since $a \mid a(rc)$ and $a \mid (bc)s$ we obtain from here that $a \mid c$. $\square$

Corollary 24.4

If F is a field then the ring of polynomials $F[x]$ is a UFD.

Proof. Follows from Theorem 24.1 and Theorem 21.13 $\square$